1-13. A core with three legs is shown in Figure P1-10. Its depth is 5 cm, and there are 400 turns on the center leg. The remaining dimensions are shown in the figure. The core is composed of a steel having the magnetization curve shown in Figure 1-10*c*. Answer the following questions about this core:

(a) What current is required to produce a flux density of 0.5 T in the central leg of the core?

(b) What current is required to produce a flux density of 1.0 T in the central leg of the core? Is it twice the current in part *(a)*?

(c) What are the reluctances of the central and right legs of the core under the conditions in part *(a)*?

(d) What are the reluctances of the central and right legs of the core under the conditions in part *(b)*?

(e) What conclusion can you make about reluctances in real magnetic cores?

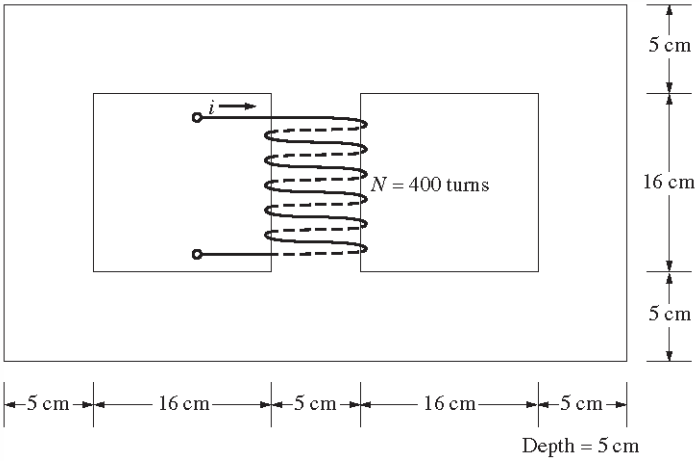
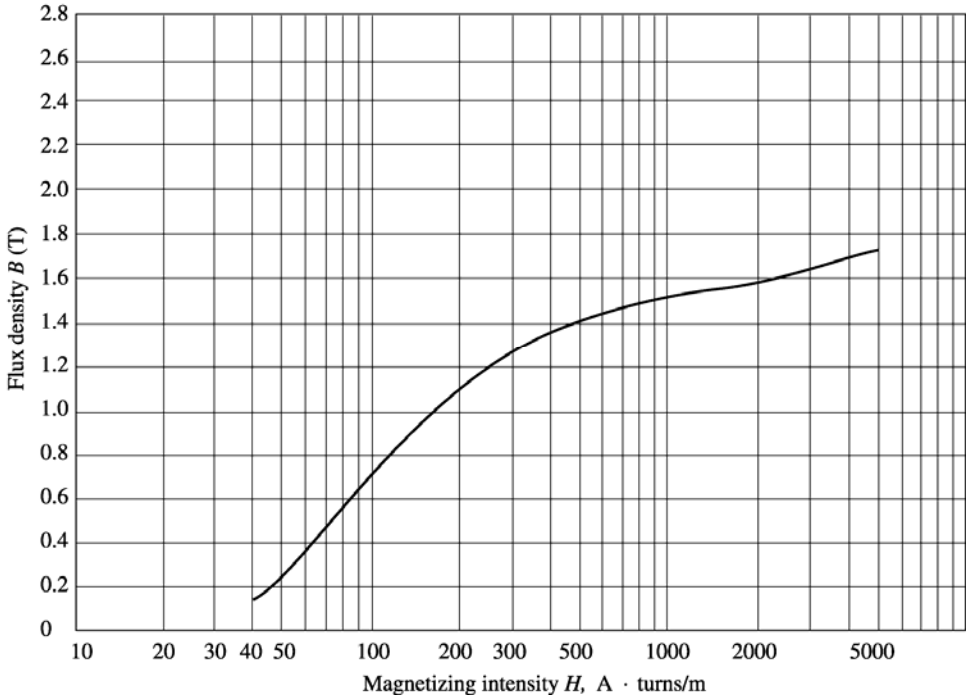


FIGURE P1–10
The core of Problem 1–13.

SOLUTION The magnetization curve for this core is shown below:



(a) A flux density of 0.5 T in the central core corresponds to a total flux of

$$\phi_{TOT} = BA = (0.5\text{ T})(0.05\text{ m})(0.05\text{ m}) = 0.00125\text{ Wb}$$

By symmetry, the flux in each of the two outer legs must be $\phi_1 = \phi_2 = 0.000625$ Wb, and the flux density in the other legs must be

$$B_1 = B_2 = \frac{0.000625\text{ Wb}}{(0.05\text{ m})(0.05\text{ m})} = 0.25\text{ T}$$

The magnetizing intensity *H* required to produce a flux density of 0.25 T can be found from Figure 1-10*c*. It is 50 A·t/m. Similarly, the magnetizing intensity *H* required to produce a flux density of 0.50 T is 75 A·t/m. The mean length of the center leg is 21 cm and the mean length of each outer leg is 63 dm, so the total MMF needed is

$$\mathcal{F}_{TOT} = H_{center}\ l_{center} + H_{outer}\ l_{outer}$$
$$\mathcal{F}_{TOT} = (75\text{ A}\cdot\text{t/m})(0.21\text{ m}) + (50\text{ A}\cdot\text{t/m})(0.63\text{ m}) = 47.3\text{ A}\cdot\text{t}$$

and the required current is

$$i = \frac{\mathcal{F}_{TOT}}{N} = \frac{47.3\text{ A}\cdot\text{t}}{400\text{ t}} = 0.12\text{ A}$$

(b) A flux density of 1.0 T in the central core corresponds to a total flux of

$$\phi_{TOT} = BA = (1.0\text{ T})(0.05\text{ m})(0.05\text{ m}) = 0.0025\text{ Wb}$$

By symmetry, the flux in each of the two outer legs must be $\phi_1 = \phi_2 = 0.00125$ Wb, and the flux density in the other legs must be

$$B_1 = B_2 = \frac{0.00125\text{ Wb}}{(0.05\text{ m})(0.05\text{ m})} = 0.50\text{ T}$$

The magnetizing intensity *H* required to produce a flux density of 0.50 T can be found from Figure 1-10*c*. It is 75 A·t/m. Similarly, the magnetizing intensity *H* required to produce a flux density of 1.00 T is about 160 A·t/m. Therefore, the total MMF needed is

$$\mathcal{F}_{TOT} = H_{center} I_{center} + H_{outer} I_{outer}$$
$$\mathcal{F}_{TOT} = (160\text{ A}\cdot\text{t/m})(0.21\text{ m}) + (75\text{ A}\cdot\text{t/m})(0.63\text{ m}) = 80.8\text{ A}\cdot\text{t}$$

and the required current is

$$i = \frac{\phi_{TOT}}{N} = \frac{80.8\text{ A}\cdot\text{t}}{400\text{ t}} = 0.202\text{ A}$$

This current is *not* twice the current in part *(a)*.

(c) The reluctance of the central leg of the core under the conditions of part *(a)* is:

$$\mathcal{R}_{cent} = \frac{\mathcal{F}_{TOT}}{\phi_{TOT}} = \frac{(75\text{ A}\cdot\text{t/m})(0.21\text{ m})}{0.00125\text{ Wb}} = 12.6\text{ kA}\cdot\text{t/Wb}$$

The reluctance of the right leg of the core under the conditions of part *(a)* is:

$$\mathcal{R}_{right} = \frac{\mathcal{F}_{TOT}}{\phi_{TOT}} = \frac{(50\text{ A}\cdot\text{t/m})(0.63\text{ m})}{0.000625\text{ Wb}} = 50.4\text{ kA}\cdot\text{t/Wb}$$

(d) The reluctance of the central leg of the core under the conditions of part *(b)* is:

$$\mathcal{R}_{cent} = \frac{\mathcal{F}_{TOT}}{\phi_{TOT}} = \frac{(160\text{ A}\cdot\text{t/m})(0.21\text{ m})}{0.0025\text{ Wb}} = 13.4\text{ kA}\cdot\text{t/Wb}$$

The reluctance of the right leg of the core under the conditions of part *(b)* is:

$$\mathcal{R}_{right} = \frac{\mathcal{F}_{TOT}}{\phi_{TOT}} = \frac{(75\text{ A}\cdot\text{t/m})(0.63\text{ m})}{0.00125\text{ Wb}} = 37.8\text{ kA}\cdot\text{t/Wb}$$

1-17. Figure P1-13 shows the core of a simple dc motor. The magnetization curve for the metal in this core is given by Figure 1-10*c* and *d*. Assume that the cross-sectional area of each air gap is 18 cm^2 and that the width of each air gap is 0.05 cm. The effective diameter of the rotor core is 5 cm.

(a) We wish to build a machine with as great a flux density as possible while avoiding excessive saturation in the core. What would be a reasonable maximum flux density for this core?

(b) What would be the total flux in the core at the flux density of part *(a)*?

(c) The maximum possible field current for this machine is 1 A. Select a reasonable number of turns of wire to provide the desired flux density while not exceeding the maximum available current.

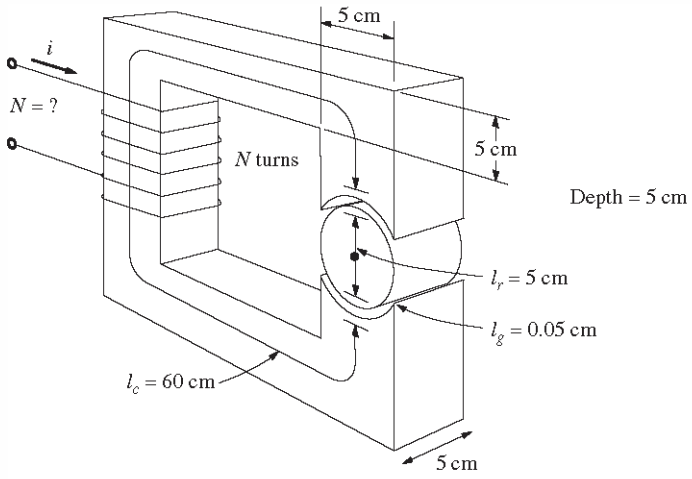
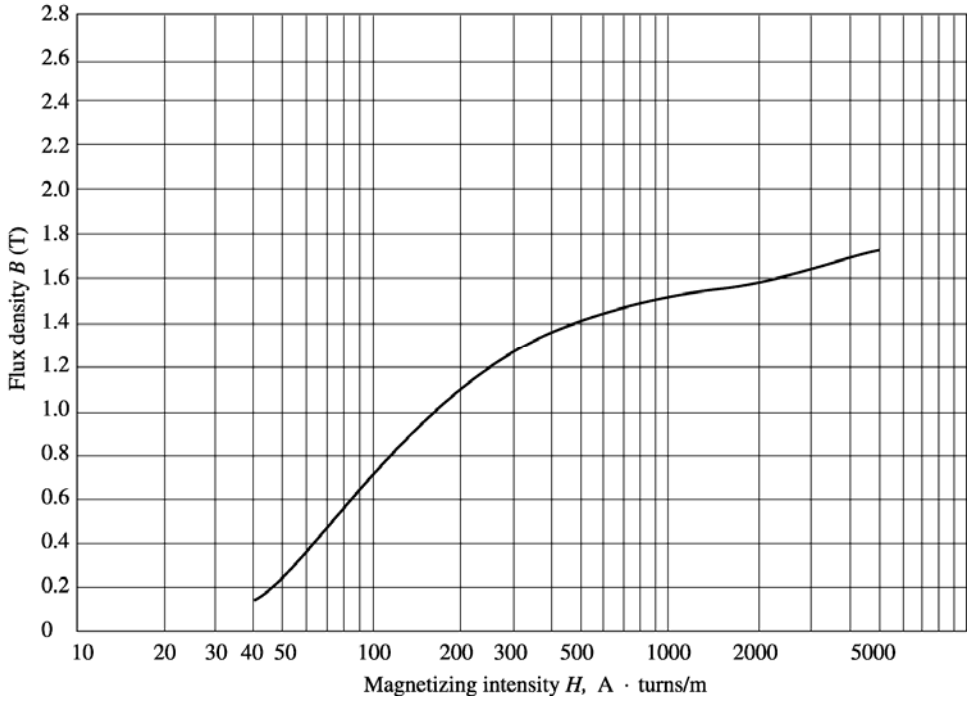
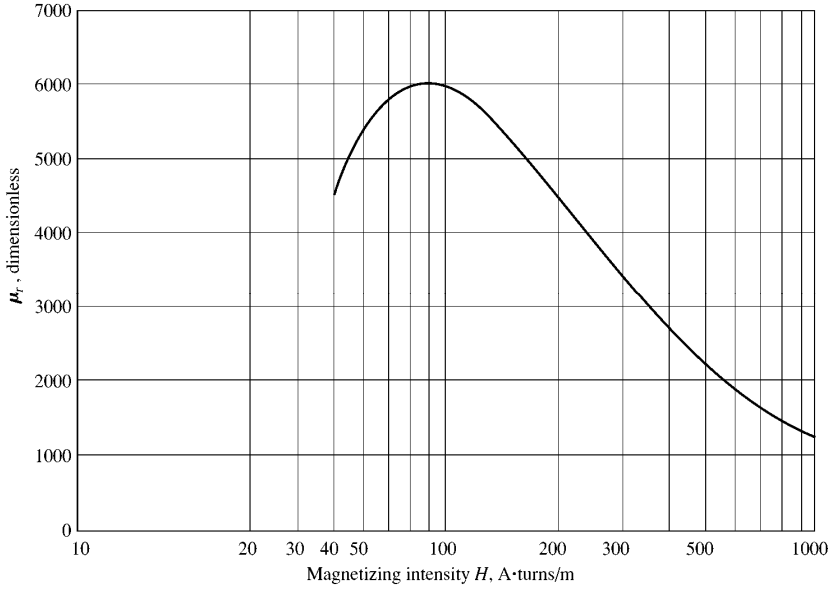


FIGURE P1–13
The core of Problem 1–17.

SOLUTION The magnetization curve for this core is shown below:



The relative permeability of this core is shown below:



Note: This is a design problem, and the answer presented here is not unique. Other values could be selected for the flux density in part *(a)*, and other numbers of turns could be selected in part *(c)*. These other answers are also correct if the proper steps were followed, and if the choices were reasonable.

(a) From Figure 1-10*c*, a reasonable maximum flux density would be about 1.2 T. Notice that the saturation effects become significant for higher flux densities.

(b) At a flux density of 1.2 T, the total flux in the core would be

$$\phi = BA = (1.2\text{ T})(0.05\text{ m})(0.05\text{ m}) = 0.0030\text{ Wb}$$

(c) The total reluctance of the core is:

$$\mathcal{R}_{\text{TOT}} = \mathcal{R}_{\text{stator}} + \mathcal{R}_{\text{air gap 1}} + \mathcal{R}_{\text{rotor}} + \mathcal{R}_{\text{air gap 2}}$$

At a flux density of 1.2 T, the relative permeability μ_r of the stator is about 3800, so the stator reluctance is

$$\mathcal{R}_{\text{stator}} = \frac{l_{\text{stator}}}{\mu_{\text{stator}} A_{\text{stator}}} = \frac{0.60 \text{ m}}{(3800)(4\pi \times 10^{-7} \text{ H/m})(0.05 \text{ m})(0.05 \text{ m})} = 50.3 \text{ kA} \cdot \text{t/Wb}$$

At a flux density of 1.2 T, the relative permeability μ_r of the rotor is 3800, so the rotor reluctance is

$$\mathcal{R}_{\text{rotor}} = \frac{l_{\text{rotor}}}{\mu_{\text{stator}} A_{\text{rotor}}} = \frac{0.05 \text{ m}}{(3800)(4\pi \times 10^{-7} \text{ H/m})(0.05 \text{ m})(0.05 \text{ m})} = 4.2 \text{ kA} \cdot \text{t/Wb}$$

The reluctance of both air gap 1 and air gap 2 is

$$\mathcal{R}_{\text{air gap 1}} = \mathcal{R}_{\text{air gap 2}} = \frac{l_{\text{air gap}}}{\mu_{\text{air gap}} A_{\text{air gap}}} = \frac{0.0005 \text{ m}}{(4\pi \times 10^{-7} \text{ H/m})(0.0018 \text{ m}^2)} = 221 \text{ kA} \cdot \text{t/Wb}$$

Therefore, the total reluctance of the core is

$$\mathcal{R}_{\text{TOT}} = \mathcal{R}_{\text{stator}} + \mathcal{R}_{\text{air gap 1}} + \mathcal{R}_{\text{rotor}} + \mathcal{R}_{\text{air gap 2}}$$

$$\mathcal{R}_{\text{TOT}} = 50.3 + 221 + 4.2 + 221 = 496 \text{ kA} \cdot \text{t/Wb}$$

The required MMF is

$$\mathcal{F}_{\text{TOT}} = \phi \mathcal{R}_{\text{TOT}} = (0.003 \text{ Wb})(496 \text{ kA} \cdot \text{t/Wb}) = 1488 \text{ A} \cdot \text{t}$$

Since $\mathcal{F} = Ni$, and the current is limited to 1 A, one possible choice for the number of turns is $N = 2000$. This would allow the desired flux density to be achieved with a current of about 0.74 A.

1-18. Assume that the voltage applied to a load is $\mathbf{V} = 208\angle -30° \text{ V}$ and the current flowing through the load is $\mathbf{I} = 2\angle 20° \text{ A}$.

(a) Calculate the complex power **S** consumed by this load.

(b) Is this load inductive or capacitive?

(c) Calculate the power factor of this load?

(d) Calculate the reactive power consumed or supplied by this load. Does the load consume reactive power from the source or supply it to the source?

SOLUTION

(a) The complex power **S** consumed by this load is

$$\mathbf{S} = \mathbf{VI}^* = (208\angle -30° \text{ V})(2\angle 20° \text{ A})^* = (208\angle -30° \text{ V})(2\angle -20° \text{ A})$$

$$\mathbf{S} = 416\angle -50° \text{ VA}$$

(b) This is a capacitive load.

(c) The power factor of this load is

$$\text{PF} = \cos(-50°) = 0.643 \text{ leading}$$

(d) This load supplies reactive power to the source. The reactive power of the load is

$$Q = VI \sin\theta = (208 \text{ V})(2 \text{ A}) \sin(-50°) = -319 \text{ var}$$

1-19. Figure P1-14 shows a simple single-phase ac power system with three loads. The voltage source is $\mathbf{V} = 240\angle 0° \text{ V}$, impedances of these three loads are

$$\mathbf{Z}_1 = 10\angle 30° \ \Omega \qquad \mathbf{Z}_2 = 10\angle 45° \ \Omega \qquad \mathbf{Z}_3 = 10\angle -90° \ \Omega$$

Answer the following questions about this power system.

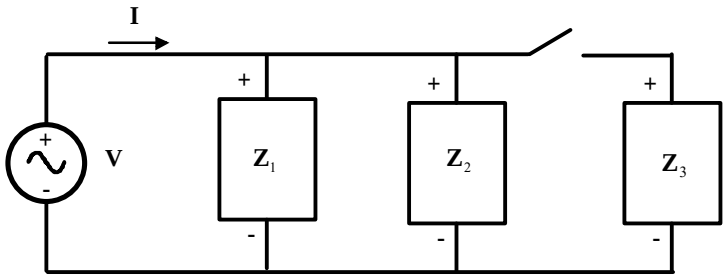
(a) Assume that the switch shown in the figure is initially open, and calculate the current **I**, the power factor, and the real, reactive, and apparent power being supplied by the source.

(b) How much real, reactive, and apparent power is being consumed by each load with the switch open?

(c) Assume that the switch shown in the figure is now closed, and calculate the current **I**, the power factor, and the real, reactive, and apparent power being supplied by the source.

(d) How much real, reactive, and apparent power is being consumed by each load with the switch closed?

(e) What happened to the current flowing from the source when the switch closed? Why?



$\mathbf{V} = 120\angle 0°\ \text{V}$

SOLUTION

(a) With the switch open, only loads 1 and 2 are connected to the source. The current $\mathbf{I}_1$ in Load 1 is

$$\mathbf{I}_1 = \frac{240\angle 0°\ \text{V}}{10\angle 30°\ \text{A}} = 24\angle -30°\ \text{A}$$

The current $\mathbf{I}_2$ in Load 2 is

$$\mathbf{I}_2 = \frac{240\angle 0°\ \text{V}}{10\angle 45°\ \text{A}} = 24\angle -45°\ \text{A}$$

Therefore the total current from the source is

$$\mathbf{I} = \mathbf{I}_1 + \mathbf{I}_2 = 24\angle -30°\ \text{A} + 24\angle -45°\ \text{A} = 47.59\angle -37.5°\ \text{A}$$

The power factor supplied by the source is

$$\text{PF} = \cos\theta = \cos(37.5°) = 0.793 \text{ lagging}$$

Note that the angle θ used in the power factor and power calculations is the *impedance angle*, which is the negative of the current angle as long as voltage is at 0°.

The real, reactive, and apparent power supplied by the source are

$$P = VI\cos\theta = (240\ \text{V})(47.59\ \text{A})\cos(37.5°) = 9061\ \text{W}$$

$$Q = VI\cos\theta = (240\ \text{V})(47.59\ \text{A})\sin(37.5°) = 6953\ \text{var}$$

$$S = VI = (240\ \text{V})(47.59\ \text{A}) = 11{,}420\ \text{VA}$$

(b) The real, reactive, and apparent power consumed by Load 1 are

$$P = VI\cos\theta = (240\ \text{V})(24\ \text{A})\cos(30°) = 4988\ \text{W}$$

$$Q = VI\cos\theta = (240\ \text{V})(24\ \text{A})\sin(30°) = 2880\ \text{var}$$

$$S = VI\cos\theta = (240\ \text{V})(24\ \text{A}) = 5760\ \text{VA}$$

The real, reactive, and apparent power consumed by Load 2 are

$$P = VI\cos\theta = (240\ \text{V})(24\ \text{A})\cos(45°) = 4073\ \text{W}$$

$$Q = VI\cos\theta = (240\text{ V})(24\text{ A})\sin(45°) = 4073\text{ var}$$
$$S = VI\cos\theta = (240\text{ V})(24\text{ A}) = 5760\text{ VA}$$

As expected, the real and reactive power supplied by the source are equal to the sum of the real and reactive powers consumed by the loads.

(c) With the switch closed, all three loads are connected to the source. The current in Loads 1 and 2 is the same as before. The current $\mathbf{I}_3$ in Load 3 is

$$\mathbf{I}_3 = \frac{240\angle 0°\text{ V}}{10\angle -90°\text{ A}} = 24\angle 90°\text{ A}$$

Therefore the total current from the source is

$$\mathbf{I} = \mathbf{I}_1 + \mathbf{I}_2 + \mathbf{I}_3 = 24\angle -30°\text{ A} + 24\angle -45°\text{ A} + 24\angle 90°\text{ A} = 38.08\angle -7.5°\text{ A}$$

The power factor supplied by the source is

$$\text{PF} = \cos\theta = \cos(7.5°) = 0.991\text{ lagging}$$

The real, reactive, and apparent power supplied by the source are

$$P = VI\cos\theta = (240\text{ V})(38.08\text{ A})\cos(7.5°) = 9061\text{ W}$$
$$Q = VI\cos\theta = (240\text{ V})(38.08\text{ A})\sin(7.5°) = 1193\text{ var}$$
$$S = VI = (240\text{ V})(38.08\text{ A}) = 9140\text{ VA}$$

(d) The real, reactive, and apparent power consumed by Load 1 are

$$P = VI\cos\theta = (240\text{ V})(24\text{ A})\cos(30°) = 4988\text{ W}$$
$$Q = VI\cos\theta = (240\text{ V})(24\text{ A})\sin(30°) = 2880\text{ var}$$
$$S = VI\cos\theta = (240\text{ V})(24\text{ A}) = 5760\text{ VA}$$

The real, reactive, and apparent power consumed by Load 2 are

$$P = VI\cos\theta = (240\text{ V})(24\text{ A})\cos(45°) = 4073\text{ W}$$
$$Q = VI\cos\theta = (240\text{ V})(24\text{ A})\sin(45°) = 4073\text{ var}$$
$$S = VI\cos\theta = (240\text{ V})(24\text{ A}) = 5760\text{ VA}$$

The real, reactive, and apparent power consumed by Load 3 are

$$P = VI\cos\theta = (240\text{ V})(24\text{ A})\cos(-90°) = 0\text{ W}$$
$$Q = VI\cos\theta = (240\text{ V})(24\text{ A})\sin(-90°) = -5760\text{ var}$$
$$S = VI\cos\theta = (240\text{ V})(24\text{ A}) = 5760\text{ VA}$$

As expected, the real and reactive power supplied by the source are equal to the sum of the real and reactive powers consumed by the loads.

(e) The current flowing *decreased* when the switch closed, because most of the reactive power being consumed by Loads 1 and 2 is being supplied by Load 3. Since less reactive power has to be supplied by the source, the total current flow decreases.

1-20. Demonstrate that Equation (1-59) can be derived from Equation (1-58) using simple trigonometric identities:

$$p(t) = v(t)\ i(t) = 2VI\cos\omega t\cos(\omega t - \theta) \tag{1-58}$$

$$p(t) = VI\cos\theta(1 + \cos 2\omega t) + VI\sin\theta\sin 2\omega t \tag{1-59}$$

SOLUTION

The first step is to apply the following identity:

$$\cos\alpha\cos\beta = \frac{1}{2}\left[\cos(\alpha-\beta)+\cos(\alpha+\beta)\right]$$

The result is

$$p(t) = v(t)\ i(t) = 2VI\cos\omega t\cos(\omega t-\theta))$$

$$p(t) = 2VI\left\{\frac{1}{2}\left[\cos(\omega t-\omega t+\theta)+\cos(\omega t+\omega t-\theta)\right]\right\}$$

$$p(t) = VI\left[\cos\theta+\cos(2\omega t-\theta)\right]$$

Now we must apply the angle addition identity to the second term:

$$\cos(\alpha-\beta) = \cos\alpha\cos\beta+\sin\alpha\sin\beta$$

The result is

$$p(t) = VI\left[\cos\theta+\cos 2\omega t\cos\theta+\sin 2\omega t\sin\theta\right]$$

Collecting terms yields the final result:

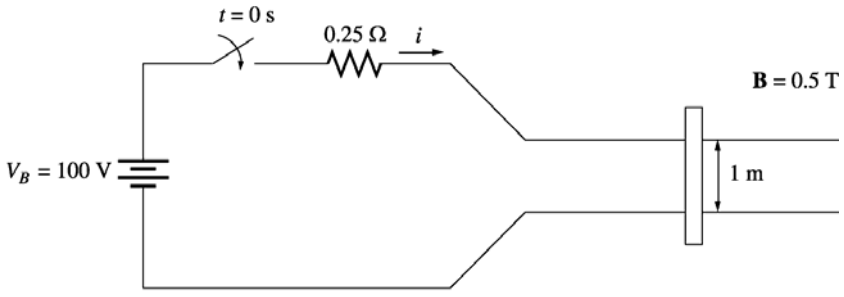
$$p(t) = VI\cos\theta(1+\cos 2\omega t)+VI\sin\theta\sin 2\omega t$$

1-21. A linear machine has a magnetic flux density of 0.5 T directed into the page, a resistance of 0.25 Ω, a bar length l = 1.0 m, and a battery voltage of 100 V.

(a) What is the initial force on the bar at starting? What is the initial current flow?

(b) What is the no-load steady-state speed of the bar?

(c) If the bar is loaded with a force of 25 N opposite to the direction of motion, what is the new steady-state speed? What is the efficiency of the machine under these circumstances?



SOLUTION

(a) The current in the bar at starting is

$$i = \frac{V_B}{R} = \frac{100\text{ V}}{0.25\ \Omega} = 400\text{ A}$$

Therefore, the force on the bar at starting is

$$\mathbf{F} = i(\mathbf{l}\times\mathbf{B}) = (400\text{ A})(1\text{ m})(0.5\text{ T}) = 200\text{ N, to the right}$$

(b) The no-load steady-state speed of this bar can be found from the equation

$$V_B = e_{\text{ind}} = vBl$$

$$v = \frac{V_B}{Bl} = \frac{100\text{ V}}{(0.5\text{ T})(1\text{ m})} = 200\text{ m/s}$$

(c) With a load of 25 N opposite to the direction of motion, the steady-state current flow in the bar will be given by

$$F_{\text{app}} = F_{\text{ind}} = ilB$$

$$i = \frac{F_{\text{app}}}{Bl} = \frac{25\text{ N}}{(0.5\text{ T})(1\text{ m})} = 50\text{ A}$$

The induced voltage in the bar will be

$$e_{\text{ind}} = V_B - iR = 100\text{ V} - (50\text{ A})(0.25\ \Omega) = 87.5\text{ V}$$

and the velocity of the bar will be

$$v = \frac{V_B}{Bl} = \frac{87.5\text{ V}}{(0.5\text{ T})(1\text{ m})} = 175\text{ m/s}$$

The *input* power to the linear machine under these conditions is

$$P_{\text{in}} = V_B i = (100\text{ V})(50\text{ A}) = 5000\text{ W}$$

The *output* power from the linear machine under these conditions is

$$P_{\text{out}} = V_B i = (87.5\text{ V})(50\text{ A}) = 4375\text{ W}$$

Therefore, the efficiency of the machine under these conditions is

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\% = \frac{4375\text{ W}}{5000\text{ W}} \times 100\% = 87.5\%$$

1-22. A linear machine has the following characteristics:

$B = 0.5$ T into page $\qquad R = 0.25\ \Omega$

$l = 0.5$ m $\qquad V_B = 120$ V

(a) If this bar has a load of 20 N attached to it opposite to the direction of motion, what is the steady-state speed of the bar?

(b) If the bar runs off into a region where the flux density falls to 0.45 T, what happens to the bar? What is its final steady-state speed?

(c) Suppose V_B is now decreased to 100 V with everything else remaining as in part *(b)*. What is the new steady-state speed of the bar?

(d) From the results for parts *(b)* and *(c)*, what are two methods of controlling the speed of a linear machine (or a real dc motor)?

SOLUTION

(a) With a load of 20 N opposite to the direction of motion, the steady-state current flow in the bar will be given by

$$F_{\text{app}} = F_{\text{ind}} = ilB$$

$$i = \frac{F_{\text{app}}}{Bl} = \frac{20\text{ N}}{(0.5\text{ T})(0.5\text{ m})} = 80\text{ A}$$

The induced voltage in the bar will be

$$e_{\text{ind}} = V_B - iR = 120 \text{ V} - (80 \text{ A})(0.25\ \Omega) = 100 \text{ V}$$

and the velocity of the bar will be

$$v = \frac{e_{\text{ind}}}{Bl} = \frac{100 \text{ V}}{(0.5 \text{ T})(0.5 \text{ m})} = 400 \text{ m/s}$$

(b) If the flux density drops to 0.45 T while the load on the bar remains the same, there will be a speed transient until $F_{\text{app}} = F_{\text{ind}} = 20 \text{ N}$ again. The new steady state current will be

$$F_{\text{app}} = F_{\text{ind}} = ilB$$

$$i = \frac{F_{\text{app}}}{Bl} = \frac{20 \text{ N}}{(0.45 \text{ T})(0.5 \text{ m})} = 88.9 \text{ A}$$

The induced voltage in the bar will be

$$e_{\text{ind}} = V_B - iR = 120 \text{ V} - (88.9 \text{ A})(0.25\ \Omega) = 97.8 \text{ V}$$

and the velocity of the bar will be

$$v = \frac{e_{\text{ind}}}{Bl} = \frac{97.8 \text{ V}}{(0.45 \text{ T})(0.5 \text{ m})} = 433 \text{ m/s}$$

(c) If the battery voltage is decreased to 100 V while the load on the bar remains the same, there will be a speed transient until $F_{\text{app}} = F_{\text{ind}} = 20 \text{ N}$ again. The new steady state current will be

$$F_{\text{app}} = F_{\text{ind}} = ilB$$

$$i = \frac{F_{\text{app}}}{Bl} = \frac{20 \text{ N}}{(0.45 \text{ T})(0.5 \text{ m})} = 88.9 \text{ A}$$

The induced voltage in the bar will be

$$e_{\text{ind}} = V_B - iR = 100 \text{ V} - (88.9 \text{ A})(0.25\ \Omega) = 77.8 \text{ V}$$

and the velocity of the bar will be

$$v = \frac{e_{\text{ind}}}{Bl} = \frac{77.8 \text{ V}}{(0.45 \text{ T})(0.5 \text{ m})} = 344 \text{ m/s}$$

(d) From the results of the two previous parts, we can see that there are two ways to control the speed of a linear dc machine. *Reducing* the flux density *B* of the machine *increases* the steady-state speed, and *reducing* the battery voltage V_B *decreases* the stead-state speed of the machine. Both of these speed control methods work for real dc machines as well as for linear machines.

Chapter 7: *DC Machinery Fundamentals*

7-1. The following information is given about the simple rotating loop shown in Figure 7-6:

$B = 0.4$ T $\qquad V_B = 48$ V

$l = 0.5$ m $\qquad R = 0.4\ \Omega$

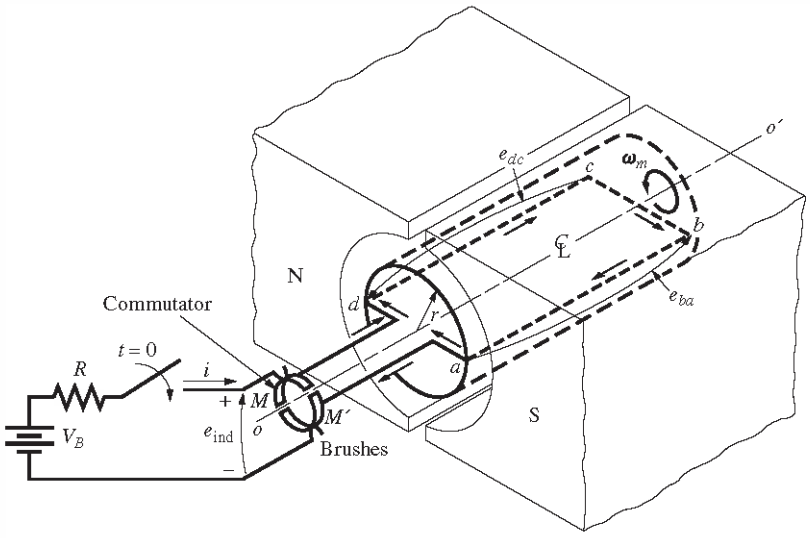
$r = 0.25$ m $\qquad \omega = 500$ rad/s

(a) Is this machine operating as a motor or a generator? Explain.

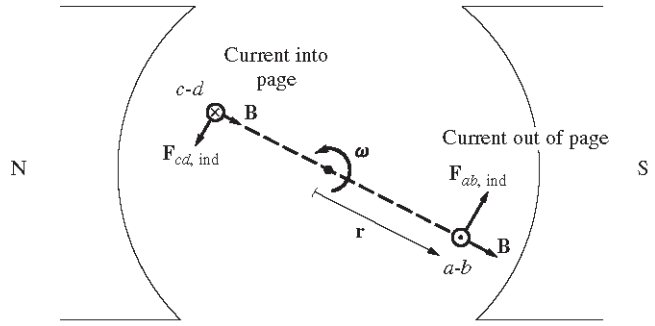
(b) What is the current i flowing into or out of the machine? What is the power flowing into or out of the machine?

(c) If the speed of the rotor were changed to 550 rad/s, what would happen to the current flow into or out of the machine?

(d) If the speed of the rotor were changed to 450 rad/s, what would happen to the current flow into or out of the machine?



(a)



(b)

FIGURE 7–6
Derivation of an equation for the induced torque in the loop. Note that the iron core is not shown in part **b** for clarity.

SOLUTION

(a) If the speed of rotation ω of the shaft is 500 rad/s, then the voltage induced in the rotating loop will be

$$e_{\text{ind}} = 2rlB\omega$$
$$e_{\text{ind}} = 2(0.25\text{ m})(0.5\text{ m})(0.4\text{ T})(500\text{ rad/s}) = 50\text{ V}$$

Since the external battery voltage is only 48 V, this machine is operating as a *generator*, charging the battery.

(b) The current flowing out of the machine is approximately

$$i = \frac{e_{\text{ind}} - V_B}{R} = \left(\frac{50\text{ V} - 48\text{ V}}{0.4\ \Omega}\right) = 5.0\text{ A}$$

(*Note* that this value is the current flowing while the loop is under the pole faces. When the loop goes beyond the pole faces, e_{ind} will momentarily fall to 0 V, and the current flow will momentarily reverse. Therefore, the *average* current flow over a complete cycle will be somewhat less than 5.0 A.)

(c) If the speed of the rotor were increased to 550 rad/s, the induced voltage of the loop would increase to

$$e_{\text{ind}} = 2rlB\omega$$
$$e_{\text{ind}} = 2(0.25\text{ m})(0.5\text{ m})(0.4\text{ T})(550\text{ rad/s}) = 55\text{ V}$$

and the current flow out of the machine will increase to

$$i = \frac{e_{\text{ind}} - V_B}{R} = \left(\frac{55\text{ V} - 48\text{ V}}{0.4\ \Omega}\right) = 17.5\text{ A}$$

(d) If the speed of the rotor were decreased to 450 rad/s, the induced voltage of the loop would fall to

$$e_{\text{ind}} = 2rlB\omega$$
$$e_{\text{ind}} = 2(0.25\text{ m})(0.5\text{ m})(0.4\text{ T})(450\text{ rad/s}) = 45\text{ V}$$

Here, e_{ind} is less than V_B, so current flows into the loop and the machine is acting as a motor. The current flow into the machine would be

$$i = \frac{V_B - e_{\text{ind}}}{R} = \left(\frac{48\text{ V} - 45\text{ V}}{0.4\ \Omega}\right) = 7.5\text{ A}$$

7-2. Refer to the simple two-pole eight-coil machine shown in Figure P7-1. The following information is given about this machine:

Figure P7-1

$B = 1.0$ T in air gap

$l = 0.3$ m (length of coil sides)

$r = 0.10$ m (radius of coils)

$n = 1800$ r/min CCW

The resistance of each rotor coil is 0.04 Ω.

(a) Is the armature winding shown a progressive or retrogressive winding?

(b) How many current paths are there through the armature of this machine?

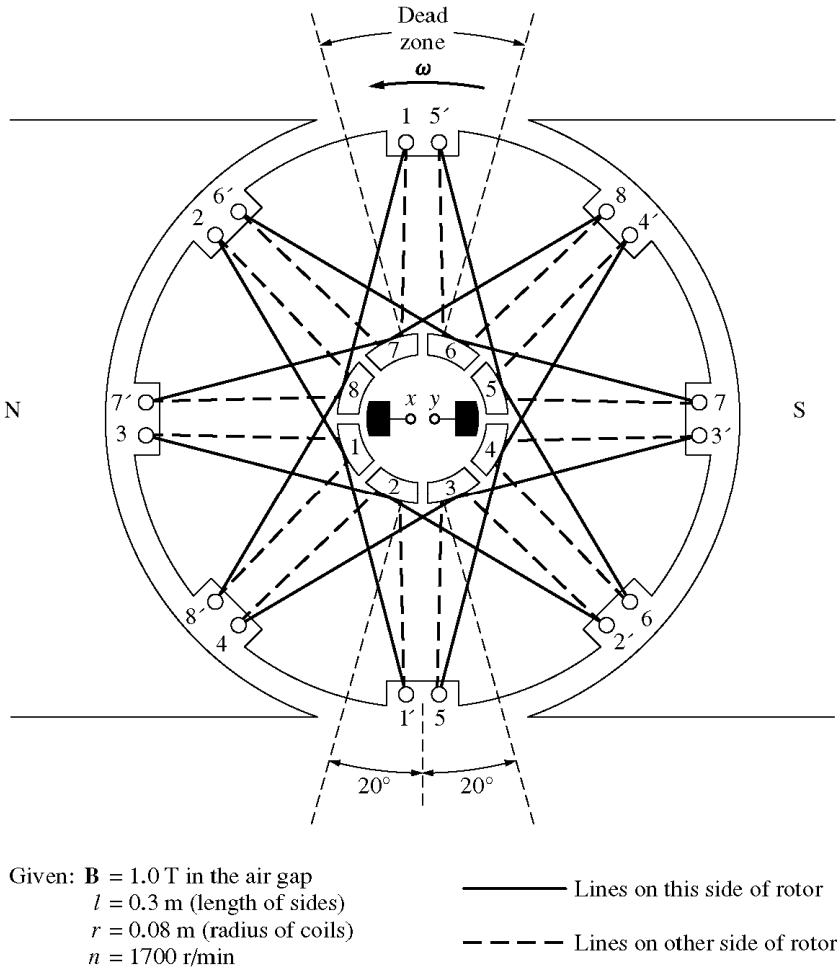
(c) What are the magnitude and the polarity of the voltage at the brushes in this machine?

(d) What is the armature resistance R_A of this machine?

(e) If a 1 Ω resistor is connected to the terminals of this machine, how much current flows in the machine? Consider the internal resistance of the machine in determining the current flow.

(f) What are the magnitude and the direction of the resulting induced torque?

(g) Assuming that the speed of rotation and magnetic flux density are constant, plot the terminal voltage of this machine as a function of the current drawn from it.



SOLUTION

(a) This winding is *progressive*, since the ends of each coil are connected to the commutator segments ahead of the segments that the beginnings of the coils are connected to.

(b) There are *two* current paths in parallel through the armature of this machine (this is a simplex lap winding).

(c) The voltage is positive at brush *x* with respect to brush *y*, since the voltage in the conductors is positive out of the page under the North pole face and positive into the page under the South pole face.

The voltage produced by this machine can be found from Equations 7-32 and 7-33:

$$E_A = \frac{ZvBl}{a} = \frac{Zr\omega Bl}{a}$$

where Z is the number of conductors *under the pole faces*, since the ones between the poles have no voltage in them. There are 16 conductors in this machine, and about 12 of them are under the pole faces at any given time.

$$\omega = (1800 \text{ r/min})\left(\frac{2\pi \text{ rad}}{1 \text{ r}}\right)\left(\frac{1 \text{ min}}{60 \text{ s}}\right) = 188.5 \text{ rad/s}$$

$$E_A = \frac{Zr\omega Bl}{a} = \frac{(12 \text{ cond})(0.10 \text{ m})(188.5 \text{ rad/s})(1.0 \text{ T})(0.3 \text{ m})}{2 \text{ current paths}} = 33.9 \text{ V}$$

(d) There are 8 coils on this machine in two parallel paths, with each coil having a resistance of 0.04 Ω. Therefore, the total resistance R_A is

$$R_A = \frac{(0.04\ \Omega\ +\ 0.04\ \Omega\ +\ 0.04\ \Omega\ +\ 0.04\ \Omega)(0.04\ \Omega\ +\ 0.04\ \Omega\ +\ 0.04\ \Omega\ +\ 0.04\ \Omega)}{0.04\ \Omega\ +\ 0.04\ \Omega\ +\ 0.04\ \Omega\ +\ 0.04\ \Omega + 0.04\ \Omega\ +\ 0.04\ \Omega\ +\ 0.04\ \Omega\ +\ 0.04\ \Omega}$$

$$R_A = 0.08\ \Omega$$

(e) The voltage produced by this machine is 33.9 V, as found in part *(c)*. Therefore, the current flowing in the machine will be

$$I_A = \frac{E_A}{R_A + R_{\text{load}}} = \frac{33.9 \text{ V}}{0.08\ \Omega + 1.0\ \Omega} = 31.4 \text{ A}$$

(f) The induced torque is given by Equation 7-46:

$$\tau_{\text{ind}} = \frac{ZrlBI_A}{a} = \frac{(12 \text{ cond})(0.10 \text{ m})(0.3 \text{ m})(1.0 \text{ T})(33.9 \text{ A})}{2 \text{ current paths}}$$

$$\tau_{\text{ind}} = 4.12 \text{ N} \cdot \text{m, CW} \quad (\textit{opposite} \text{ to the direction of rotation})$$

(e) The terminal voltage of this machine is given by

$$V_T = E_A - I_A R_A$$

```
% M-file: prob7_2.m
% M-file to create a plot of the terminal voltage vs load
% current for the machine of problem 7-2.

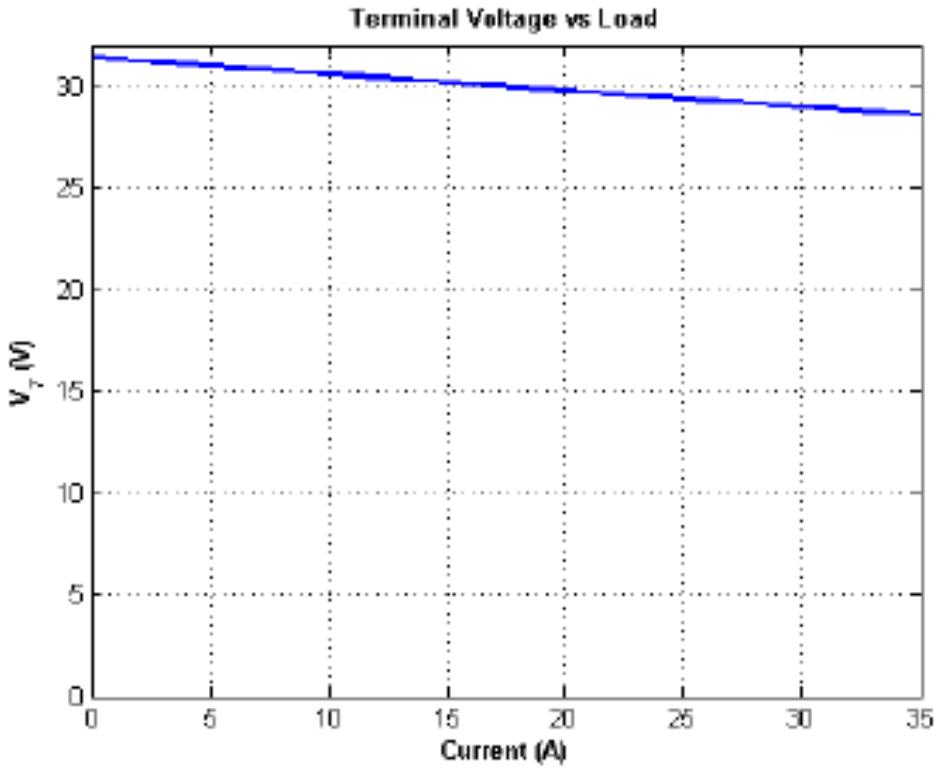
% First, initialize the values needed in this program.
Ea = 31.4;                  % Stator resistance
Ra = 0.08;                  % Stator reactance

% Get the line currents to calculate the anser for
Ia = 0:1:35;

% Calculate the terminal voltage
Vt = Ea - Ia * Ra;

% Plot the terminal voltage
figure(1);
plot(Ia,Vt,'b-','LineWidth',2.0);
xlabel('\bfCurrent (A)');
ylabel('\bf\itV_{T} \rm\bf(V)');
title ('\bfTerminal Voltage vs Load');
axis([ 0 35 0 32]);
grid on;
hold off;
```

The resulting plot is shown below.



7-3. Prove that the equation for the induced voltage of a single simple rotating loop

$$e_{\text{ind}} = \frac{2}{\pi} \phi \omega \tag{7-6}$$

is just a special case of the general equation for induced voltage in a dc machine

$$E_A = K \phi \omega \tag{7-38}$$

SOLUTION From Equation 7-38,

$$E_A = K \phi \omega$$

where $K = \dfrac{ZP}{2\pi a}$

For the simple rotation loop,

$Z = 2$ (There are 2 conductors)
$P = 2$ (There are 2 poles)
$a = 1$ (There is one current path through the machine)

Therefore,

$$K = \frac{ZP}{2\pi a} \frac{(2)\ (2)}{2\pi\ (1)} = \frac{2}{\pi}$$

$a = 2m = (2)(4) = 8$ paths

7-6. The power converted from one form to another within a dc motor was given by

$$P_{\text{conv}} = E_A I_A = \tau_{\text{ind}} \omega_m$$

Use the equations for E_A and τ_{ind} [Equations (7-38) and (7-49)] to prove that $E_A I_A = \tau_{\text{ind}} \omega_m$; that is, prove that the electric power disappearing at the point of power conversion is exactly equal to the mechanical power appearing at that point.

SOLUTION

$$P_{\text{conv}} = E_A I_A$$

Substituting Equation (8-38) for E_A

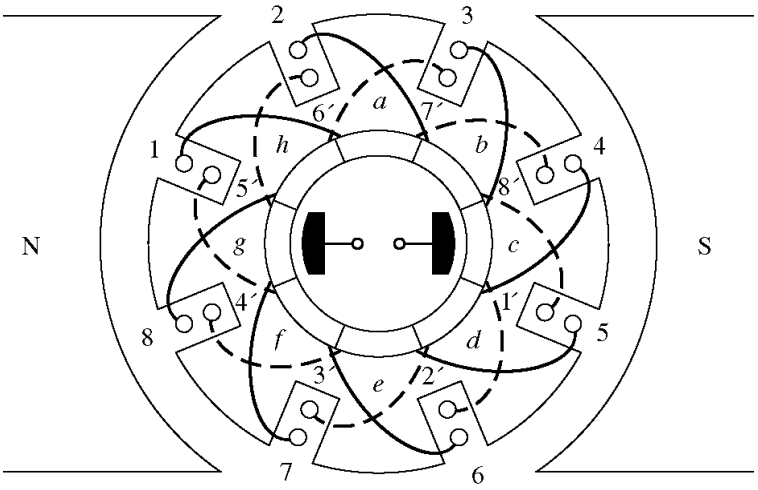
$$P_{\text{conv}} = (K \phi \omega) I_A$$

$$P_{\text{conv}} = (K \phi I_A) \omega$$

But from Equation (7-49), $\tau_{\text{ind}} = K \phi I_A$, so

$$P_{\text{conv}} \tau_{\text{ind}} \omega$$

7-8. Figure P7-2 shows a small two-pole dc motor with eight rotor coils and 10 turns per coil. The flux per pole in this machine is 0.006 Wb.



(a) If this motor is connected to a 12-V dc car battery, what will the no-load speed of the motor be?

(b) If the positive terminal of the battery is connected to the rightmost brush on the motor, which way will it rotate?

(c) If this motor is loaded down so that it consumes 600 W from the battery, what will the induced torque of the motor be? (Ignore any internal resistance in the motor.)

SOLUTION

(a) At no load, $V_T = E_A = K\phi\omega$. If K is known, then the speed of the motor can be found. The constant K is given by

$$K = \frac{ZP}{2\pi a}$$

On the average, about 6 of the 8 coils are under the pole faces at any given time, so the average number of active conductors is

$$Z = (6 \text{ coils})(4 \text{ turns/coil})(2 \text{ conductors/turn}) = 48 \text{ conductors}$$

There are two poles and two current paths, so

$$K = \frac{ZP}{2\pi a} = \frac{(48 \text{ cond})(2 \text{ poles})}{2\pi(2 \text{ paths})} = 7.64$$

The speed is given by

$$\omega = \frac{E_A}{K\phi} = \frac{12 \text{ V}}{(7.64)(0.006 \text{ Wb})} = 262 \text{ rad/s}$$

$$n_m = (262 \text{ rad/s})\left(\frac{1 \text{ r}}{2\pi \text{ rad}}\right)\left(\frac{60 \text{ s}}{1 \text{ min}}\right) = 2500 \text{ r/min}$$

(b) If the positive terminal of the battery is connected to the rightmost brush, current will flow into the page under the South pole face, producing a CW torque $\Rightarrow$ CW rotation.

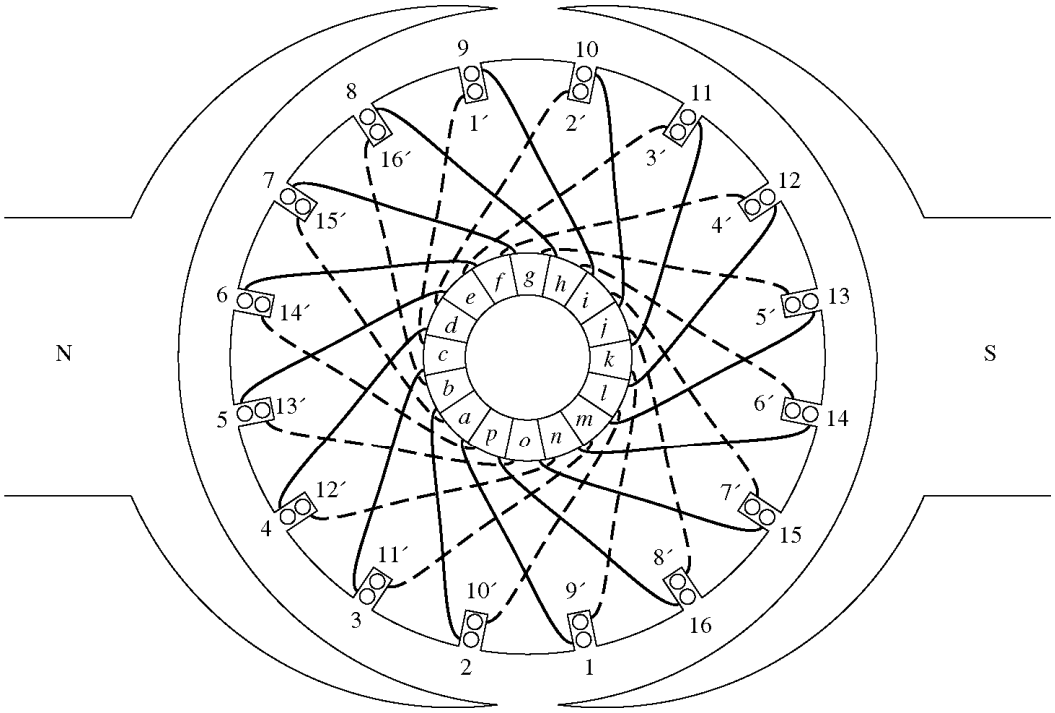
(c) If the motor consumes 600 W from the battery, the current flow is

$$I = \frac{P}{V_B} = \frac{600 \text{ W}}{12 \text{ V}} = 50 \text{ A}$$

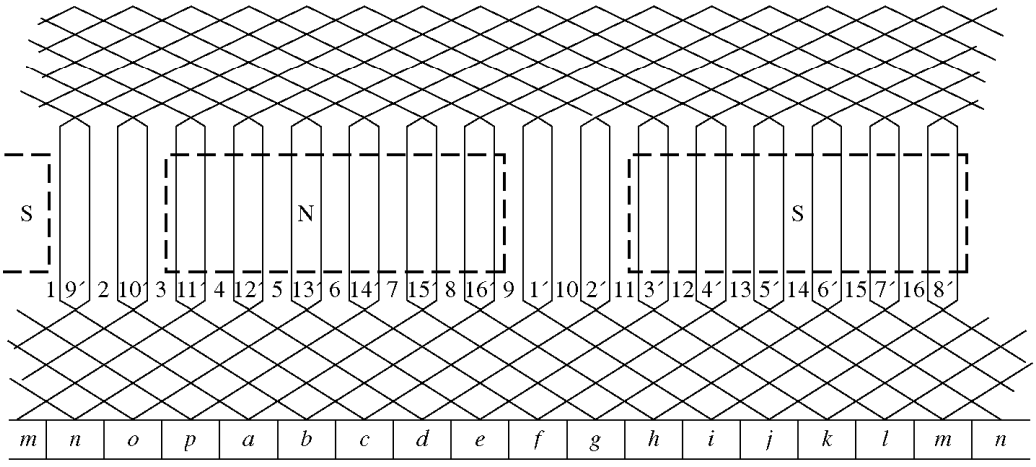
Therefore, the induced torque will be

$$\tau_{\text{ind}} = K\phi I_A = (7.64)(0.006 \text{ Wb})(50 \text{ A}) = 2.29 \text{ N} \cdot \text{m, CW}$$

7-9. Refer to the machine winding shown in Figure P7-3.



(*a*)



(*b*)

(a) How many parallel current paths are there through this armature winding?

(b) Where should the brushes be located on this machine for proper commutation? How wide should they be?

(c) What is the plex of this machine?

(d) If the voltage on any single conductor under the pole faces in this machine is *e*, what is the voltage at the terminals of this machine?

SOLUTION

(a) This is a *duplex*, *two-pole*, *lap winding*, so there are 4 parallel current paths through the rotor.

(b) The brushes should be shorting out those windings lying between the two poles. At the time shown, those windings are 1, 2, 9, and 10. Therefore, the brushes should be connected to short out commutator segments *b*-*c*-*d* and *j*-*k*-*l* at the instant shown in the figure. Each brush should be two commutator segments wide, since this is a duplex winding.

(c) Duplex (see above)

(d) There are 16 coils on the armature of this machine. Of that number, an average of 14 of them would be under the pole faces at any one time. Therefore, there are 28 conductors divided among 4 parallel paths, which produces 7 conductors per path. Therefore, $E_A = 7e = V_T$ for no-load conditions.

8-9. For the separately excited motor of Problem 8-8:

(a) What is the maximum no-load speed attainable by varying both V_A and R_{adj}?

(b) What is the minimum no-load speed attainable by varying both V_A and R_{adj}?

(c) What is the motor's efficiency at rated conditions? [*Note:* Assume that (1) the brush voltage drop is 2 V; (2) the core loss is to e determined at an armature voltage equal to the armature voltage under full load; and (3) stray load losses are 1 percent of full load.]

SOLUTION

(a) The maximum speed will occur with the maximum V_A and the maximum R_{adj}. The field current when $R_{\text{adj}} = 400\ \Omega$ is:

$$I_F = \frac{V_T}{R_{\text{adj}} + R_F} = \frac{240\text{ V}}{400\ \Omega + 75\ \Omega} = \frac{240\text{ V}}{475\ \Omega} = 0.505\text{ A}$$

From Figure P8-1, this field current would produce an internal generated voltage E_{Ao} of 167 V at a speed n_o of 1200 r/min. At no-load conditions, the maximum internal generated voltage $E_A = V_A$ = 240 V. Therefore, the speed *n* with a voltage of 240 V would be

$$\frac{E_A}{E_{Ao}} = \frac{n}{n_o}$$

$$n = \left(\frac{E_A}{E_{Ao}}\right) n_o = \left(\frac{240\text{ V}}{167\text{ V}}\right)(1200\text{ r/min}) = 1725\text{ r/min}$$

(b) The minimum speed will occur with the minimum V_A and the minimum R_{adj}. The field current when $R_{\text{adj}} = 100\ \Omega$ is:

$$I_F = \frac{V_T}{R_{\text{adj}} + R_F} = \frac{240\text{ V}}{100\ \Omega + 75\ \Omega} = \frac{240\text{ V}}{175\ \Omega} = 1.37\text{ A}$$

From Figure P8-1, this field current would produce an internal generated voltage E_{Ao} of 271 V at a speed n_o of 1200 r/min. At no-load conditions, the minimum internal generated voltage $E_A = V_A$ = 120 V. Therefore, the speed *n* with a voltage of 120 V would be

$$\frac{E_A}{E_{Ao}} = \frac{n}{n_o}$$

$$n = \left(\frac{E_A}{E_{Ao}}\right) n_o = \left(\frac{120\text{ V}}{271\text{ V}}\right)(1200\text{ r/min}) = 531\text{ r/min}$$

For Problems 8-10 to 8-11, the motor is connected cumulatively compounded as shown in Figure P8-4.

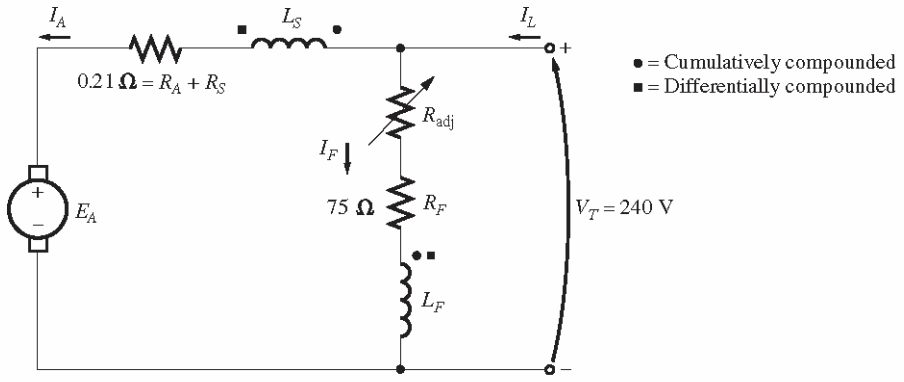


FIGURE P8–4
The equivalent circuit of the compounded motor in Problems 8–10 to 8–12.

8-10. If the motor is connected cumulatively compounded with R_{adj} = 175 Ω:

(a) What is the no-load speed of the motor?

(b) What is the full-load speed of the motor?

(c) What is its speed regulation?

(d) Calculate and plot the torque-speed characteristic for this motor. (Neglect armature effects in this problem.)

SOLUTION *At no-load conditions*, $E_A = V_T = 240$ V. The field current is given by

$$I_F = \frac{V_T}{R_{\text{adj}} + R_F} = \frac{240 \text{ V}}{175\ \Omega + 75\ \Omega} = \frac{240 \text{ V}}{250\ \Omega} = 0.960 \text{ A}$$

From Figure P8-1, this field current would produce an internal generated voltage E_{Ao} of 241 V at a speed n_o of 1200 r/min. Therefore, the speed *n* with a voltage of 240 V would be

$$\frac{E_A}{E_{Ao}} = \frac{n}{n_o}$$

$$n = \left(\frac{E_A}{E_{Ao}}\right) n_o = \left(\frac{240 \text{ V}}{241 \text{ V}}\right)(1200 \text{ r/min}) = 1195 \text{ r/min}$$

At full load conditions, the armature current is

$$I_A = I_L - I_F = I_L - \frac{V_T}{R_{\text{adj}} + R_F} = 110 \text{ A} - 0.96 \text{ A} = 109 \text{ A}$$

The internal generated voltage E_A is

$$E_A = V_T - I_A(R_A + R_S) = 240 \text{ V} - (109 \text{ A})(0.21\ \Omega) = 217.1 \text{ V}$$

The equivalent field current is

$$I_F{}^* = I_F + \frac{N_{\text{SE}}}{N_F} I_A = 0.96 \text{ A} + \frac{14 \text{ turns}}{2700 \text{ turns}}(109 \text{ A}) = 1.53 \text{ A}$$

From Figure P8-1, this field current would produce an internal generated voltage E_{Ao} of 279 V at a speed n_o of 1200 r/min. Therefore,

$$n = \left(\frac{E_A}{E_{Ao}}\right) n_o = \left(\frac{217.1 \text{ V}}{279 \text{ V}}\right)(1200 \text{ r/min}) = 934 \text{ r/min}$$

The speed regulation is

$$\text{SR} = \frac{n_{\text{nl}} - n_{\text{fl}}}{n_{\text{fl}}} \times 100\% = \frac{1195 \text{ r/min} - 934 \text{ /min}}{934 \text{ r/min}} \times 100\% = 27.9\%$$

The torque-speed characteristic can best be plotted with a MATLAB program. An appropriate program is shown below.

Compare this torque-speed curve to that of the shunt motor in Problem 8-7. (Both curves are plotted on the same scale to facilitate comparison.)

8-11. The motor is connected cumulatively compounded and is operating at full load. What will the new speed of the motor be if R_{adj} is increased to 250 Ω? How does the new speed compared to the full-load speed calculated in Problem 8-10?

SOLUTION If R_{adj} is increased to 250 Ω, the field current is given by

$$I_F = \frac{V_T}{R_{\text{adj}} + R_F} = \frac{240\text{ V}}{250\ \Omega + 75\ \Omega} = \frac{240\text{ V}}{250\ \Omega} = 0.739\text{ A}$$

At full load conditions, the armature current is

$$I_A = I_L - I_F = 110\text{ A} - 0.739\text{ A} = 109.2\text{ A}$$

The internal generated voltage E_A is

$$E_A = V_T - I_A(R_A + R_S) = 240\text{ V} - (109.2\text{ A})(0.21\ \Omega) = 217.1\text{ V}$$

The equivalent field current is

$$I_F^* = I_F + \frac{N_{\text{SE}}}{N_F} I_A = 0.739\text{ A} + \frac{14\text{ turns}}{2700\text{ turns}}(109.2\text{ A}) = 1.305\text{ A}$$

From Figure P9-1, this field current would produce an internal generated voltage E_{Ao} of 268 V at a speed n_o of 1200 r/min. Therefore,

$$n = \left(\frac{E_A}{E_{Ao}}\right) n_o = \left(\frac{217.1\text{ V}}{268\text{ V}}\right)(1200\text{ r/min}) = 972\text{ r/min}$$

The new full-load speed is higher than the full-load speed in Problem 8-10.

8-13. A **7.5**-hp 120-V series dc motor has an armature resistance of 0.1 Ω and a series field resistance of 0.08 Ω. At full load, the current input is **56** A, and the rated speed is 1050 r/min. Its magnetization curve is shown in Figure P8-5. The core losses are **220** W, and the mechanical losses are **230** W at full load. Assume that the mechanical losses vary as the cube of the speed of the motor and that the core losses are constant.

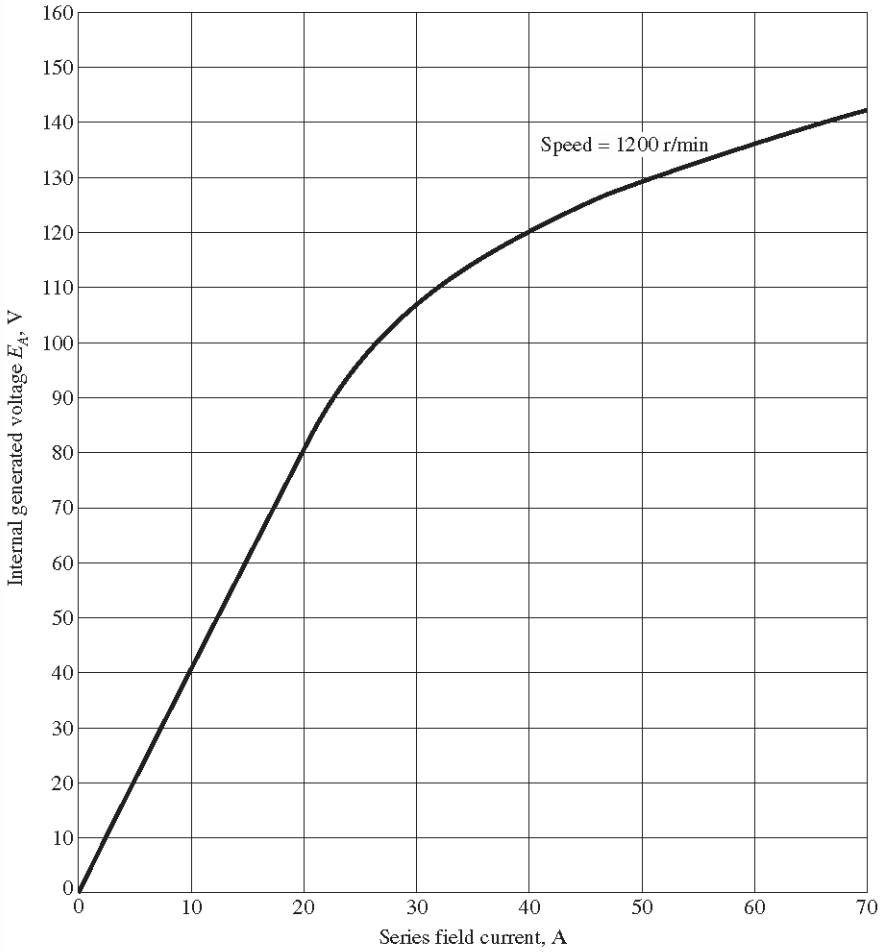


FIGURE P8–5
The magnetization curve for the series motor in Problem 8–13. This curve was taken at a constant speed of 1200 r/min.

(a) What is the efficiency of the motor at full load?

(b) What are the speed and efficiency of the motor if it is operating at an armature current of **40** A?

(c) Plot the torque-speed characteristic for this motor.

Note: An electronic version of this magnetization curve can be found in file `p85_mag.dat`, which can be used with MATLAB programs. Column 1 contains field current in amps, and column 2 contains the internal generated voltage E_A in volts.

SOLUTION

(a) The output power of this motor at full load is

$$P_{\text{OUT}} = (7.5 \text{ hp})(746 \text{ W/hp}) = 5595 \text{ W}$$

The input power is

$$P_{\text{IN}} = V_T I_L = (120 \text{ V})(56 \text{ A}) = 6720 \text{ W}$$

Therefore the efficiency is

$$\eta = \frac{P_{\text{OUT}}}{P_{\text{IN}}} \times 100\% = \frac{5595 \text{ W}}{6720 \text{ W}} \times 100\% = 83.3\%$$

(b) If the armature current is 40 A, then the input power to the motor will be

$$P_{\text{IN}} = V_T I_L = (120 \text{ V})(40 \text{ A}) = 4800 \text{ W}$$

The internal generated voltage at this condition is

$$E_{A2} = V_T - I_A (R_A + R_S) = 120 \text{ V} - (40 \text{ A})(0.10 \ \Omega + 0.08 \ \Omega) = 112.8 \text{ V}$$

and the internal generated voltage at rated conditions is

$$E_{A2} = V_T - I_A (R_A + R_S) = 120 \text{ V} - (56 \text{ A})(0.10 \ \Omega + 0.08 \ \Omega) = 109.9 \text{ V}$$

The final speed is given by the equation

$$\frac{E_{A2}}{E_{A1}} = \frac{K \ \phi_2 \ \omega_2}{K \ \phi_2 \ \omega_2} = \frac{E_{Ao,2} \ n_2}{E_{Ao,1} \ n_1}$$

since the ratio $E_{Ao,2} / E_{Ao,1}$ is the same as the ratio ϕ_2 / ϕ_1. Therefore, the final speed is

$$n_2 = \frac{E_{A2}}{E_{A1}} \frac{E_{Ao,1}}{E_{Ao,2}} n_1$$

From Figure P8-5, the internal generated voltage $E_{Ao,2}$ for a current of 40 A and a speed of n_o = 1200 r/min is $E_{Ao,2}$ = 120 V, and the internal generated voltage $E_{Ao,1}$ for a current of 56 A and a speed of n_o = 1200 r/min is $E_{Ao,1}$ = 133 V.

$$n_2 = \frac{E_{A2}}{E_{A1}} \frac{E_{Ao,1}}{E_{Ao,2}} n_1 = \left(\frac{112.8 \text{ V}}{109.9 \text{ V}}\right)\left(\frac{133 \text{ V}}{120 \text{ V}}\right)(1050 \text{ r/min}) = 1195 \text{ r/min}$$

The power converted from electrical to mechanical form is

$$P_{\text{conv}} = E_A I_A = (112.8 \text{ V})(40 \text{ A}) = 4512 \text{ W}$$

The core losses in the motor are 220 W, and the mechanical losses in the motor are 230 W at a speed of 1050 r/min. The mechanical losses in the motor scale proportionally to the cube of the rotational speedm so the mechanical losses at 1326 r/min are

$$P_{\text{mech}} = \left(\frac{n_2}{n_1}\right)^3 (230 \text{ W}) = \left(\frac{1195 \text{ r/min}}{1050 \text{ r/min}}\right)^3 (230 \text{ W}) = 339 \text{ W}$$

Therefore, the output power is

$$P_{\text{OUT}} = P_{\text{conv}} - P_{\text{mech}} - P_{\text{core}} = 4512 \text{ W} - 339 \text{ W} - 220 \text{ W} = 3953 \text{ W}$$

and the efficiency is

$$\eta = \frac{P_{\text{OUT}}}{P_{\text{IN}}} \times 100\% = \frac{3953 \text{ W}}{4800 \text{ W}} \times 100\% = 82.4\%$$

8-14. A 20-hp 240-V 76-A 900 r/min series motor has a field winding of 33 turns per pole. Its armature resistance is 0.09 Ω, and its field resistance is 0.06 Ω. The magnetization curve expressed in terms of magnetomotive force versus E_A at 900 r/min is given by the following table:

E_A, V	95	150	188	212	229	243
$\mathcal{F}$, A·turns	500	1000	1500	2000	2500	3000

Armature reaction is negligible in this machine.

(a) Compute the motor's torque, speed, and output power at 33, 67, 100, and 133 percent of full-load armature current. (Neglect rotational losses.)

(b) Plot the terminal characteristic of this machine.

Note: An electronic version of this magnetization curve can be found in file `prob8_14_mag.dat`, which can be used with MATLAB programs. Column 1 contains magnetomotive force in ampere-turns, and column 2 contains the internal generated voltage E_A in volts.

SOLUTION Note that this magnetization curve has been stored in a file called `prob8_14_mag.dat`. The first column of the file is an array of `mmf_values`, and the second column is an array of `ea_values`. These values are valid at a speed n_o = 900 r/min. Because the data in the file is relatively sparse, it is

(a) Since full load corresponds to 76 A, this calculation must be performed for armature currents of 25.3 A, 50.7 A, 76 A, and 101.3 A.

If I_A = 23.3 A, then

$$E_A = V_T - I_A(R_A + R_S) = 240\text{ V} - (25.3\text{ A})(0.09\ \Omega + 0.06\ \Omega) = 236.2\text{ V}$$

The magnetomotive force is $\mathcal{F} = NI_A = (33\text{ turns})(25.3\text{ A}) = 835\text{ A}\cdot\text{turns}$, which produces a voltage E_{Ao} of 134 V at n_o = 900 r/min. Therefore the speed of the motor at these conditions is

$$n = \frac{E_A}{E_{Ao}} n_o = \frac{236.2\text{ V}}{134\text{ V}}(900\text{ r/min}) = 1586\text{ r/min}$$

The power converted from electrical to mechanical form is

$$P_{\text{conv}} = E_A I_A = (236.2\text{ V})(25.3\text{ A}) = 5976\text{ W}$$

Since the rotational losses are ignored, this is also the output power of the motor. The induced torque is

$$\tau_{\text{ind}} = \frac{P_{\text{conv}}}{\omega_m} = \frac{5976\text{ W}}{(1586\text{ r/min})\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right)} = 36\text{ N}\cdot\text{m}$$

If I_A = 50.7 A, then

$$E_A = V_T - I_A\left(R_A + R_S\right) = 240\text{ V} - \left(50.7\text{ A}\right)\left(0.09\ \Omega + 0.06\ \Omega\right) = 232.4\text{ V}$$

The magnetomotive force is $\mathcal{F} = NI_A = \left(33\text{ turns}\right)\left(50.7\text{ A}\right) = 1672\text{ A}\cdot\text{turns}$, which produces a voltage E_{Ao} of 197 V at n_o = 900 r/min. Therefore the speed of the motor at these conditions is

$$n = \frac{E_A}{E_{Ao}} n_o = \frac{232.4\text{ V}}{197\text{ V}}\left(900\text{ r/min}\right) = 1062\text{ r/min}$$

The power converted from electrical to mechanical form is

$$P_{\text{conv}} = E_A I_A = \left(232.4\text{ V}\right)\left(50.7\text{ A}\right) = 11{,}780\text{ W}$$

Since the rotational losses are ignored, this is also the output power of the motor. The induced torque is

$$\tau_{\text{ind}} = \frac{P_{\text{conv}}}{\omega_m} = \frac{11{,}780\text{ W}}{\left(1062\text{ r/min}\right)\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right)} = 106\text{ N}\cdot\text{m}$$

If I_A = 76 A, then

$$E_A = V_T - I_A\left(R_A + R_S\right) = 240\text{ V} - \left(76\text{ A}\right)\left(0.09\ \Omega + 0.06\ \Omega\right) = 228.6\text{ V}$$

The magnetomotive force is $\mathcal{F} = NI_A = \left(33\text{ turns}\right)\left(76\text{ A}\right) = 2508\text{ A}\cdot\text{turns}$, which produces a voltage E_{Ao} of 229 V at n_o = 900 r/min. Therefore the speed of the motor at these conditions is

$$n = \frac{E_A}{E_{Ao}} n_o = \frac{228.6\text{ V}}{229\text{ V}}\left(900\text{ r/min}\right) = 899\text{ r/min}$$

The power converted from electrical to mechanical form is

$$P_{\text{conv}} = E_A I_A = \left(228.6\text{ V}\right)\left(76\text{ A}\right) = 17{,}370\text{ W}$$

Since the rotational losses are ignored, this is also the output power of the motor. The induced torque is

$$\tau_{\text{ind}} = \frac{P_{\text{conv}}}{\omega_m} = \frac{17{,}370\text{ W}}{\left(899\text{ r/min}\right)\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right)} = 185\text{ N}\cdot\text{m}$$

If I_A = 101.3 A, then

$$E_A = V_T - I_A\left(R_A + R_S\right) = 240\text{ V} - \left(101.3\text{ A}\right)\left(0.09\ \Omega + 0.06\ \Omega\right) = 224.8\text{ V}$$

The magnetomotive force is $\mathcal{F} = NI_A = \left(33\text{ turns}\right)\left(101.3\text{ A}\right) = 3343\text{ A}\cdot\text{turns}$, which produces a voltage E_{Ao} of 252 V at n_o = 900 r/min. Therefore the speed of the motor at these conditions is

$$n = \frac{E_A}{E_{Ao}} n_o = \frac{224.8\text{ V}}{252\text{ V}}\left(900\text{ r/min}\right) = 803\text{ r/min}$$

The power converted from electrical to mechanical form is

$$P_{\text{conv}} = E_A I_A = \left(224.8\text{ V}\right)\left(101.3\text{ A}\right) = 22{,}770\text{ W}$$

Since the rotational losses are ignored, this is also the output power of the motor. The induced torque is

$$\tau_{\text{ind}} = \frac{P_{\text{conv}}}{\omega_m} = \frac{22{,}770\text{ W}}{\left(803\text{ r/min}\right)\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right)} = 271\text{ N}\cdot\text{m}$$

8-15. A 300-hp 440-V 560-A, 863 r/min shunt dc motor has been tested, and the following data were taken:

Blocked-rotor test:

$V_A = 14.9$ V exclusive of brushes $\qquad V_F = 440$ V

$I_A = 500$ A $\qquad I_F = 7.52$ A

No-load operation:

$V_A = 440$ V including brushes $\qquad I_F = 7.50$ A

$I_A = 23.1$ A $\qquad n = 863$ r/min

What is this motor's efficiency at the rated conditions? [*Note*: Assume that (1) the brush voltage drop is 2 V; (2) the core loss is to be determined at an armature voltage equal to the armature voltage under full load; and (3) stray load losses are 1 percent of full load.]

SOLUTION The armature resistance of this motor is

$$R_A = \frac{V_{A,\text{br}}}{I_{A,\text{br}}} = \frac{14.9 \text{ V}}{500 \text{ A}} = 0.0298 \ \Omega$$

Under no-load conditions, the core and mechanical losses taken together (that is, the *rotational losses*) of this motor are equal to the product of the internal generated voltage E_A and the armature current I_A, since this is no output power from the motor at no-load conditions. Therefore, the rotational losses at rated speed can be found as

$$E_A = V_A - V_{\text{brush}} - I_A R_A = 440 \text{ V} - 2 \text{ V} - (23.1 \text{ A})(0.0298 \ \Omega) = 437.3 \text{ V}$$

$$P_{\text{rot}} = P_{\text{conv}} = E_A I_A = (437.3 \text{ V})(23.1 \text{ A}) = 10.1 \text{ kW}$$

The input power to the motor *at full load* is

$$P_{\text{IN}} = V_T I_L = (440 \text{ V})(560 \text{ A}) = 246.4 \text{ kW}$$

The output power from the motor at full load is

$$P_{\text{OUT}} = P_{\text{IN}} - P_{\text{CU}} - P_{\text{rot}} - P_{\text{brush}} - P_{\text{stray}}$$

The copper losses are

$$P_{\text{CU}} = I_A^2 R_A + V_F I_F = (560 \text{ A})^2 (0.0298 \ \Omega) + (440 \text{ V})(7.52 \text{ A}) = 12.65 \text{ kW}$$

The brush losses are

$$P_{\text{brush}} = V_{\text{brush}} I_A = (2 \text{ V})(560 \text{ A}) = 1.12 \text{ kW}$$

Therefore,

$$P_{\text{OUT}} = P_{\text{IN}} - P_{\text{CU}} - P_{\text{rot}} - P_{\text{brush}} - P_{\text{stray}}$$

$$P_{\text{OUT}} = 246.4 \text{ kW} - 14.1 \text{ kW} - 12.65 \text{ kW} - 10.1 \text{ kW} - 1.12 \text{ kW} - 2.46 \text{ kW} = 206 \text{ kW}$$

The motor's efficiency at full load is

$$\eta = \frac{P_{\text{OUT}}}{P_{\text{IN}}} \times 100\% = \frac{206 \text{ kW}}{246.4 \text{ kW}} \times 100\% = 83.6\%$$

Problems 8-16 to 8-19 refer to a 240-V 100-A dc motor which has both shunt and series windings. Its characteristics are

$R_A = 0.14 \ \Omega$ $\qquad N_F = 1500$ turns

$R_S = 0.05 \ \Omega$ $\qquad N_{\text{SE}} = 15$ turns

$R_F = 200 \ \Omega$ $\qquad n_m = 3000$ r/min

$R_{\text{adj}} = 0$ to 300 Ω, currently set to 120 Ω

This motor has compensating windings and interpoles. The magnetization curve for this motor at 3000 r/min is shown in Figure P8-6.

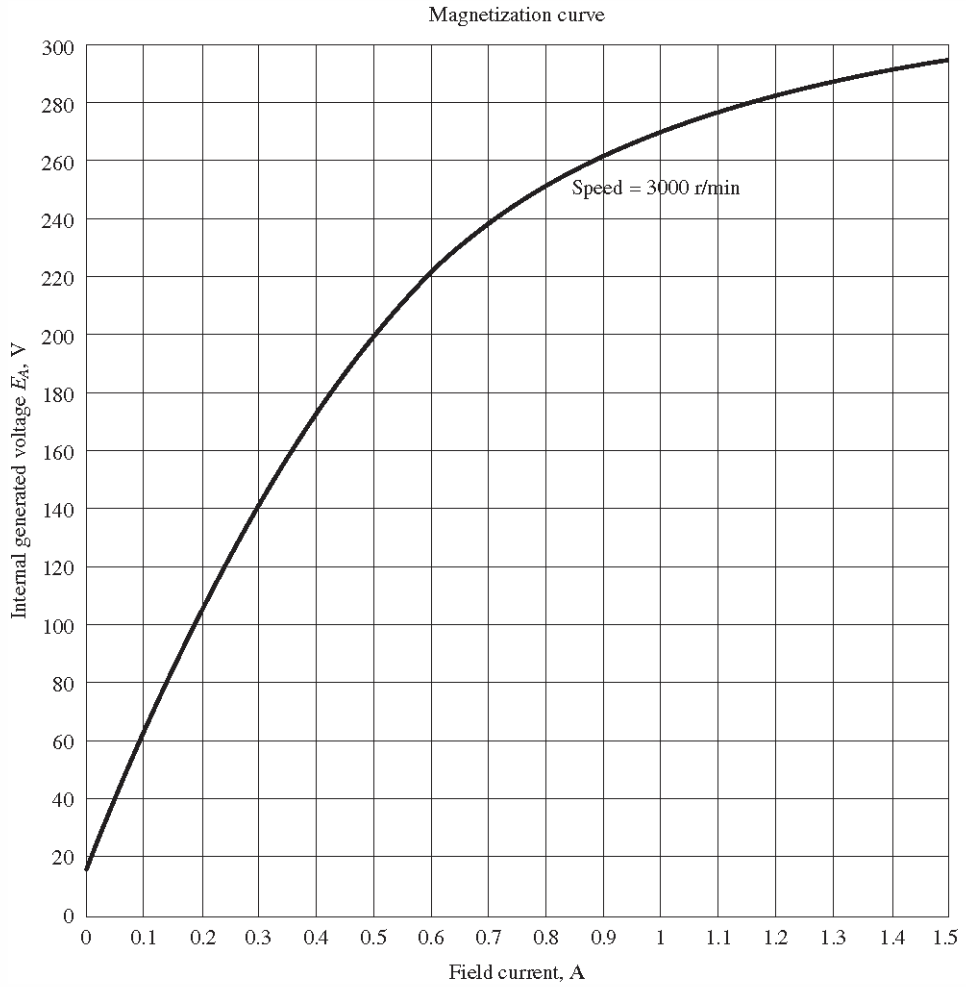


FIGURE P8–6
The magnetization curve for the dc motor in Problems 8–16 to 8–19. This curve was made at a constant speed of 3000 r/min.

Note: An electronic version of this magnetization curve can be found in file `p86_mag.dat`, which can be used with MATLAB programs. Column 1 contains field current in amps, and column 2 contains the internal generated voltage E_A in volts.

8-16. The motor described above is connected in *shunt*.

(a) What is the no-load speed of this motor when R_{adj} = 120 Ω?

(b) What is its full-load speed?

(c) What is its speed regulation?

(d) Plot the torque-speed characteristic for this motor.

(e) Under no-load conditions, what range of possible speeds can be achieved by adjusting R_{adj}?

SOLUTION Note that this magnetization curve has been stored in a file called `p86_mag.dat`. The first column of the file is an array of `ia_values`, and the second column is an array of `ea_values`. These values are valid at a speed n_o = 3000 r/min. These values can be used with the MATLAB `interp1` function to look up an internal generated voltage as follows:

(a) If R_{adj} = 120 Ω, the total field resistance is 320 Ω, and the resulting field current is

$$I_F = \frac{V_T}{R_F + R_{\text{adj}}} = \frac{240\text{ V}}{200\ \Omega\ +\ 120\ \Omega} = 0.75\text{ A}$$

This field current would produce a voltage E_{Ao} of 245 V at a speed of n_o = 3000 r/min. The actual E_A is 240 V, so the actual speed will be

$$n = \frac{E_A}{E_{Ao}} n_o = \frac{240\text{ V}}{245\text{ V}}\ (3000\text{ r/min}) = 2939\ \text{ r/min}$$

(b) At full load, $I_A = I_L - I_F = 100\text{ A} - 0.75\text{ A} = 99.25\text{ A}$, and

$$E_A = V_T - I_A R_A = 240\text{ V} - (99.25\text{ A})(0.14\ \Omega) = 226.1\text{ V}$$

Therefore, the speed at full load will be

$$n = \frac{E_A}{E_{Ao}} n_o = \frac{226.1\text{ V}}{245\text{ V}}\ (3000\text{ r/min}) = 2769\text{ r/min}$$

(c) The speed regulation of this motor is

$$\text{SR} = \frac{n_{\text{nl}} - n_{\text{fl}}}{n_{\text{fl}}} \times 100\% = \frac{2939\text{ r/min} - 2769\text{ r/min}}{2769\text{ r/min}} \times 100\% = 6.16\%$$

(e) If R_{adj} is maximum at no-load conditions, the total resistance is 500 Ω, and

$$I_F = \frac{V_T}{R_F + R_{\text{adj}}} = \frac{240\text{ V}}{200\ \Omega\ +\ 300\ \Omega} = 0.48\text{ A}$$

This field current would produce a voltage E_{Ao} of 195 V at a speed of n_o = 3000 r/min. The actual E_A is 240 V, so the actual speed will be

$$n = \frac{E_A}{E_{Ao}} n_o = \frac{240\text{ V}}{195\text{ V}}\ (3000\text{ r/min}) = 3692\text{ r/min}$$

If R_{adj} is minimum at no-load conditions, the total resistance is 200 Ω, and

$$I_F = \frac{V_T}{R_F + R_{\text{adj}}} = \frac{240\text{ V}}{200\ \Omega\ +\ 0\ \Omega} = 1.2\text{ A}$$

This field current would produce a voltage E_{Ao} of 282 V at a speed of n_o = 3000 r/min. The actual E_A is 240 V, so the actual speed will be

$$n = \frac{E_A}{E_{Ao}} n_o = \frac{240\text{ V}}{282\text{ V}}\ (3000\text{ r/min}) = 2553\text{ r/min}$$

8-21. A 10-hp 120-V 1000 r/min shunt dc motor has a full-load armature current of 70 A when operating at rated conditions. The armature resistance of the motor is R_A = 0.12 Ω, and the field resistance R_F is 40 Ω. The adjustable resistance in the field circuit R_{adj} may be varied over the range from 0 to 200 Ω and is currently set to 100 Ω. Armature reaction may be ignored in this machine. The magnetization curve for this motor, taken at a speed of 1000 r/min, is given in tabular form below:

E_A, V	5	78	95	112	118	126	130
I_F, A	0.00	0.80	1.00	1.28	1.44	2.88	4.00

(a) What is the speed of this motor when it is running at the rated conditions specified above?

(b) The output power from the motor is 10 hp at rated conditions. What is the output torque of the motor?

(c) What are the copper losses and rotational losses in the motor at full load (ignore stray losses)?

(d) What is the efficiency of the motor at full load?

(e) If the motor is now unloaded with no changes in terminal voltage or R_{adj}, what is the no-load speed of the motor?

(f) Suppose that the motor is running at the no-load conditions described in part *(e)*. What would happen to the motor if its field circuit were to open? Ignoring armature reaction, what would the final steady-state speed of the motor be under those conditions?

(g) What range of no-load speeds is possible in this motor, given the range of field resistance adjustments available with R_{adj}?

Note: An electronic version of this magnetization curve can be found in file `prob8_21_mag.dat`, which can be used with MATLAB programs. Column 1 contains field current in amps, and column 2 contains the internal generated voltage E_A in volts.

SOLUTION

(a) If R_{adj} = 100 Ω, the total field resistance is 140 Ω, and the resulting field current is

$$I_F = \frac{V_T}{R_F + R_{adj}} = \frac{120\text{ V}}{100\ \Omega\ +\ 40\ \Omega} = 0.857\text{ A}$$

This field current would produce a voltage E_{Ao} of 82.8 V at a speed of n_o = 1000 r/min. The actual E_A is

$$E_A = V_T - I_A R_A = 120\text{ V} - (70\text{ A})(0.12\ \Omega) = 111.6\text{ V}$$

so the actual speed will be

$$n = \frac{E_A}{E_{Ao}} n_o = \frac{111.6\text{ V}}{82.8\text{ V}}(1000\text{ r/min}) = 1348\text{ r/min}$$

(b) The output power is 10 hp and the output speed is 1000 r/min at rated conditions, therefore, the torque is

$$\tau_{out} = \frac{P_{out}}{\omega_m} = \frac{(10\text{ hp})(746\text{ W/hp})}{(1000\text{ r/min})\left(\frac{2\pi\text{ rad}}{1\text{ r}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right)} = 71.2\text{ N}\cdot\text{m}$$

(c) The copper losses are

$$P_{\text{CU}} = I_A{}^2 R_A + V_F I_F = (70\text{ A})^2 (0.12\ \Omega) + (120\text{ V})(0.857\text{ A}) = 691\text{ W}$$

The power converted from electrical to mechanical form is

$$P_{\text{conv}} = E_A I_A = (111.6\text{ V})(70\text{ A}) = 7812\text{ W}$$

The output power is

$$P_{\text{OUT}} = (10\text{ hp})(746\text{ W/hp}) = 7460\text{ W}$$

Therefore, the rotational losses are

$$P_{\text{rot}} = P_{\text{conv}} - P_{\text{OUT}} = 7812\text{ W} - 7460\text{ W} = 352\text{ W}$$

(d) The input power to this motor is

$$P_{\text{IN}} = V_T (I_A + I_F) = (120\text{ V})(70\text{ A} + 0.857\text{ A}) = 8503\text{ W}$$

Therefore, the efficiency is

$$\eta = \frac{P_{\text{OUT}}}{P_{\text{IN}}} \times 100\% = \frac{7460\text{ W}}{8503\text{ W}} \times 100\% = 87.7\%$$

(e) The no-load E_A will be 120 V, so the no-load speed will be

$$n = \frac{E_A}{E_{Ao}} n_o = \frac{120\text{ V}}{82.8\text{ V}} (1000\text{ r/min}) = 1450\text{ r/min}$$

(f) If the field circuit opens, the field current would go to zero $\Rightarrow \phi$ drops to $\phi_{\text{res}} \Rightarrow E_A \downarrow \Rightarrow I_A \uparrow \Rightarrow \tau_{\text{ind}} \uparrow \Rightarrow n \uparrow$ to a *very* high speed. If I_F = 0 A, E_{Ao} = 8.5 V at 1800 r/min, so

$$n = \frac{E_A}{E_{Ao}} n_o = \frac{230\text{ V}}{5\text{ V}} (1000\text{ r/min}) = 46{,}000\text{ r/min}$$

(In reality, the motor speed would be limited by rotational losses, or else the motor will destroy itself first.)

(g) The maximum value of R_{adj} = 200 Ω, so

$$I_F = \frac{V_T}{R_F + R_{\text{adj}}} = \frac{120\text{ V}}{200\ \Omega + 40\ \Omega} = 0.500\text{ A}$$

This field current would produce a voltage E_{Ao} of 50.6 V at a speed of n_o = 1000 r/min. The actual E_A is 120 V, so the actual speed will be

$$n = \frac{E_A}{E_{Ao}} n_o = \frac{120\text{ V}}{50.6\text{ V}} (1000\text{ r/min}) = 2372\text{ r/min}$$

The minimum value of R_{adj} = 0 Ω, so

$$I_F = \frac{V_T}{R_F + R_{\text{adj}}} = \frac{120\text{ V}}{0\ \Omega + 40\ \Omega} = 3.0\text{ A}$$

This field current would produce a voltage E_{Ao} of about 126.4 V at a speed of n_o = 1000 r/min. The actual E_A is 120 V, so the actual speed will be

$$n = \frac{E_A}{E_{Ao}} n_o = \frac{120\text{ V}}{126.4\text{ V}} (1000\text{ r/min}) = 949\text{ r/min}$$

8-22. The magnetization curve for a separately excited dc generator is shown in Figure P8-7. The generator is rated at 6 kW, 120 V, 50 A, and 1800 r/min and is shown in Figure P8-8. Its field circuit is rated at 5A. The following data are known about the machine:

$$R_A = 0.18\,\Omega \qquad V_F = 120\text{ V}$$
$$R_{\text{adj}} = 0 \text{ to } 40\,\Omega \qquad R_F = 20\,\Omega$$
$$N_F = 1000 \text{ turns per pole}$$

Answer the following questions about this generator, assuming no armature reaction.

(a) If this generator is operating at no load, what is the range of voltage adjustments that can be achieved by changing R_{adj}?

(b) If the field rheostat is allowed to vary from 0 to 30 Ω and the generator's speed is allowed to vary from 1500 to 2000 r/min, what are the maximum and minimum no-load voltages in the generator?

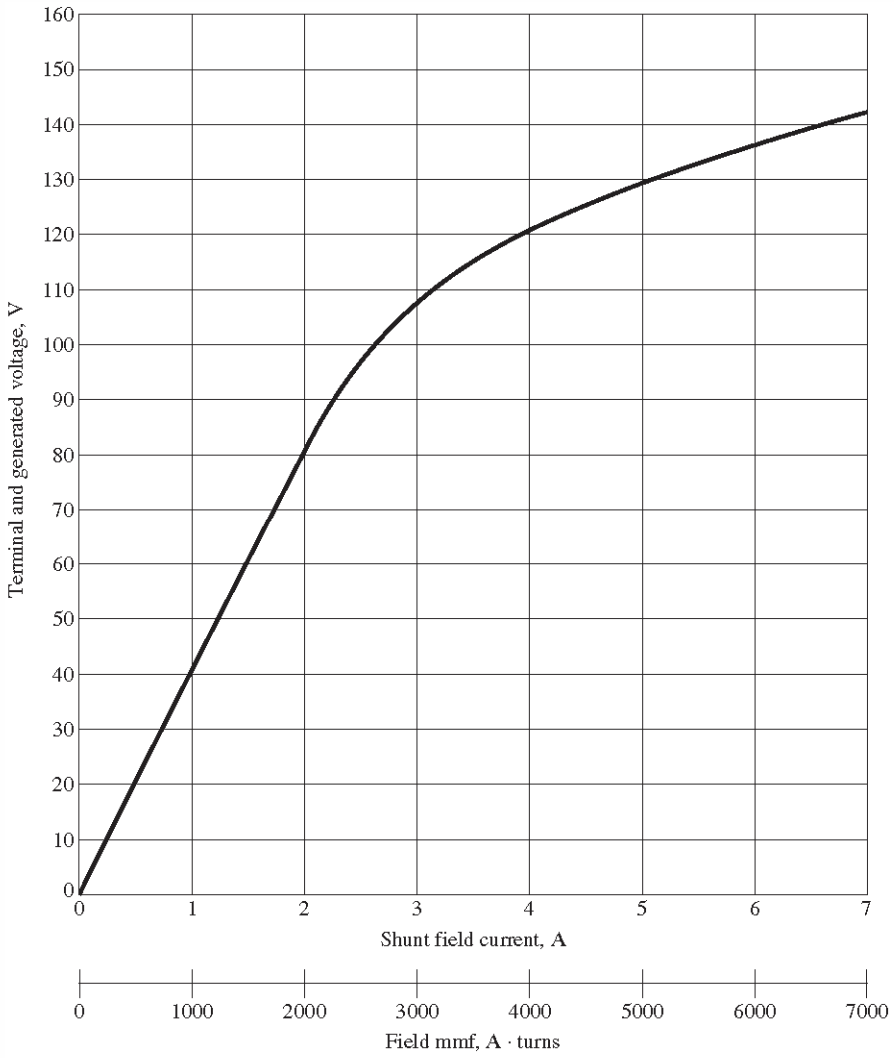


FIGURE P8–7
The magnetization curve for Problems 8–22 to 8–28. This curve was taken at a speed of 1800 r/min.

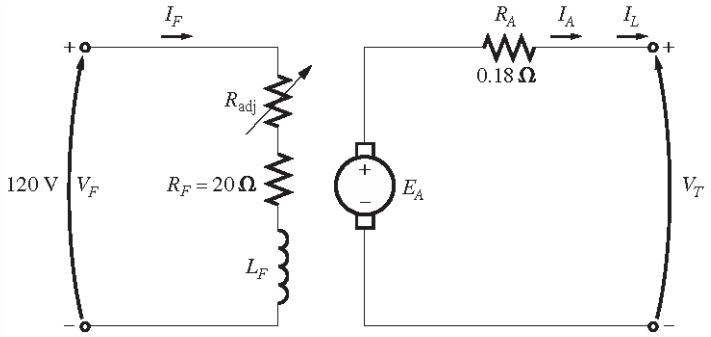


FIGURE P8–8
The separately excited dc generator in Problems 8–22 to 8–24.

SOLUTION

(a) If the generator is operating with no load at 1800 r/min, then the terminal voltage will equal the internal generated voltage E_A. The maximum possible field current occurs when $R_{\text{adj}} = 0\ \Omega$. The current is

$$I_{F,\max} = \frac{V_F}{R_F + R_{\text{adj}}} = \frac{120\text{ V}}{20\ \Omega + 0\ \Omega} = 6\text{ A}$$

From the magnetization curve, the voltage E_{Ao} at 1800 r/min is 135 V. Since the actual speed is 1800 r/min, the maximum no-load voltage is 135 V.

The minimum possible field current occurs when $R_{\text{adj}} = 40\ \Omega$. The current is

$$I_{F,\max} = \frac{V_F}{R_F + R_{\text{adj}}} = \frac{120\text{ V}}{20\ \Omega + 40\ \Omega} = 2.0\text{ A}$$

From the magnetization curve, the voltage E_{Ao} at 1800 r/min is 79.5 V. Since the actual speed is 1800 r/min, the minimum no-load voltage is 79.5 V.

(b) The maximum voltage will occur at the highest current and speed, and the minimum voltage will occur at the lowest current and speed. The maximum possible field current occurs when $R_{\text{adj}} = 0\ \Omega$. The current is

$$I_{F,\max} = \frac{V_F}{R_F + R_{\text{adj}}} = \frac{120\text{ V}}{20\ \Omega + 0\ \Omega} = 6\text{ A}$$

From the magnetization curve, the voltage E_{Ao} at 1800 r/min is 135 V. Since the actual speed is 2000 r/min, the maximum no-load voltage is

$$\frac{E_A}{E_{Ao}} = \frac{n}{n_o}$$

$$E_A = \frac{n}{n_o} E_{Ao} = \frac{2000\text{ r/min}}{1800\text{ r/min}}(135\text{ V}) = 150\text{ V}$$

The minimum possible field current occurs and minimum speed and field current. The maximum adjustable resistance is $R_{\text{adj}} = 30\ \Omega$. The current is

$$I_{F,\max} = \frac{V_F}{R_F + R_{\text{adj}}} = \frac{120\text{ V}}{20\ \Omega + 30\ \Omega} = 2.4\text{ A}$$

From the magnetization curve, the voltage E_{Ao} at 1800 r/min is 93.1 V. Since the actual speed is 1500 r/min, the maximum no-load voltage is

$$\frac{E_A}{E_{Ao}} = \frac{n}{n_o}$$

$$E_A = \frac{n}{n_o} E_{Ao} = \frac{1500\text{ r/min}}{1800\text{ r/min}}(93.1\text{ V}) = 77.6\text{ V}$$

8-25. The machine in Problem 8-22 is reconnected as a shunt generator and is shown in Figure P8-9. The shunt field resistor R_{adj} is adjusted to 10 Ω, and the generator's speed is 1800 r/min.

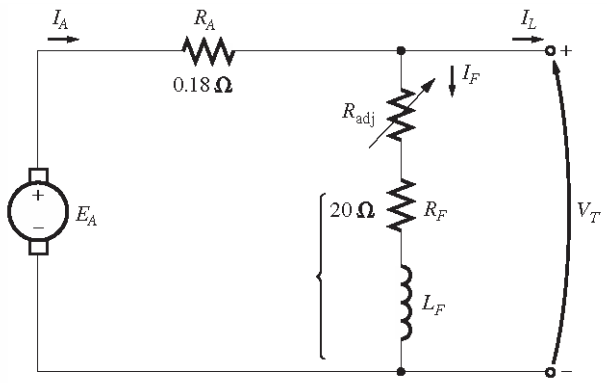


FIGURE P8–9
The shunt dc generator in Problems 8–25 and 8–26.

(a) What is the no-load terminal voltage of the generator?

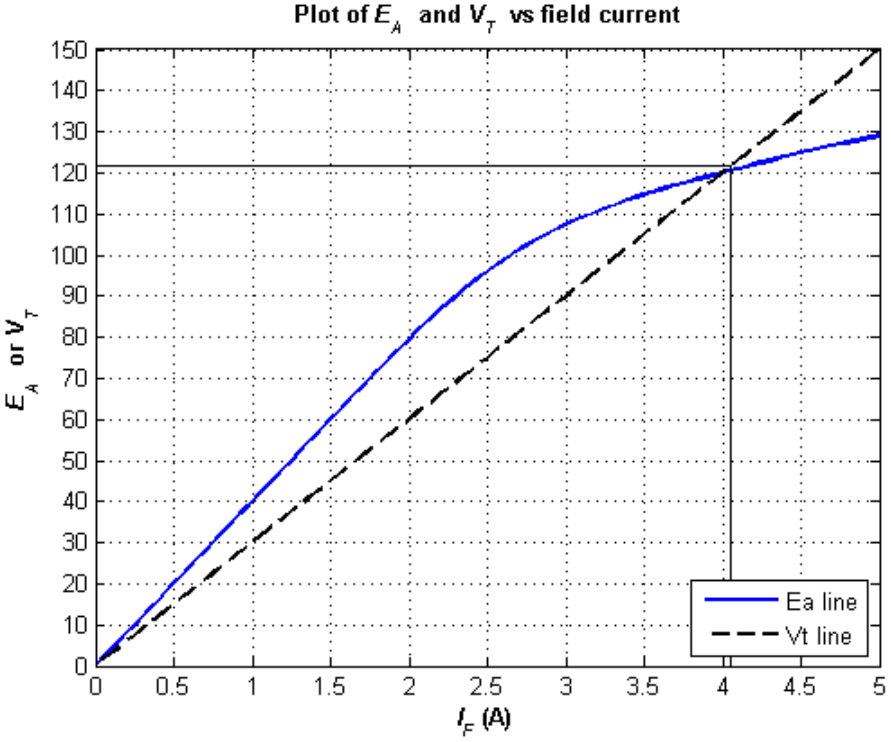
(b) Assuming no armature reaction, what is the terminal voltage of the generator with an armature current of 20 A? 40 A?

(c) Assuming an armature reaction equal to 300 A·turns at full load, what is the terminal voltage of the generator with an armature current of 20 A? 40 A?

(d) Calculate and plot the terminal characteristics of this generator with and without armature reaction.

SOLUTION

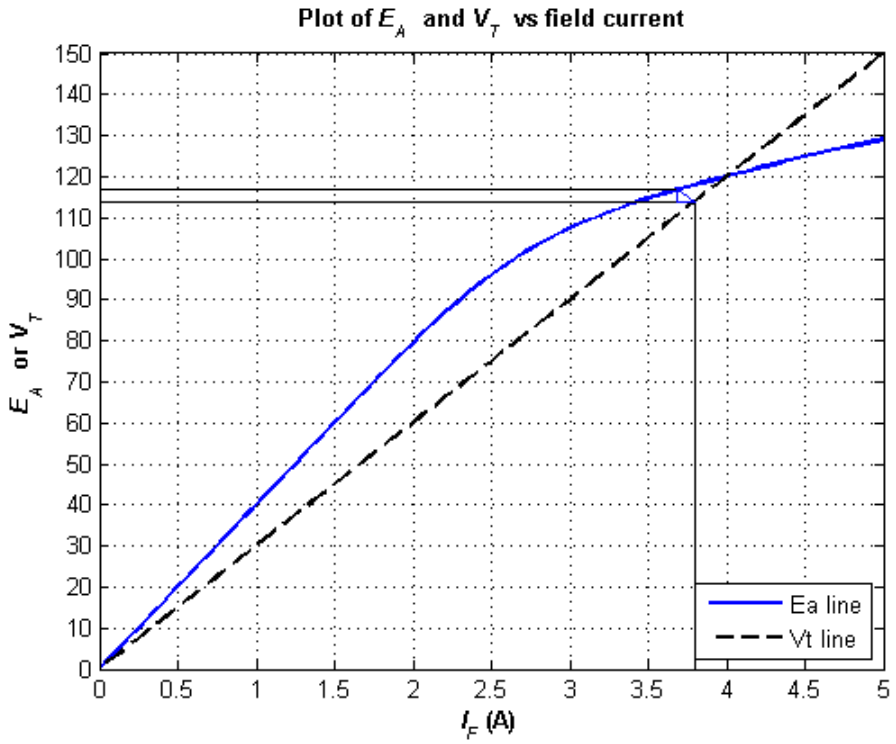
(a) The total field resistance of this generator is 30 Ω, and the no-load terminal voltage can be found from the intersection of the resistance line with the magnetization curve for this generator. The magnetization curve and the field resistance line are plotted below. As you can see, they intersect at a terminal voltage of 121 V.



(c) The rated current of this generator is 50 A, so 20 A is 40% of full load. If the full load armature reaction is 300 A·turns, and if the armature reaction is assumed to change linearly with armature current, then the armature reaction will be 120 A·turns. The demagnetizing effect of armature reaction is equivalent to a reduction in field current of

$$\Delta I_F = \frac{120 \text{ A•t}}{1000 \text{ t}} = 0.12 \text{ A}$$

The figure below shows that a triangle consisting of 3.6 V and (120 A·turns)/(1000 turns) = 0.12 A fits exactly between the E_A and V_T lines at a terminal voltage of 114 V.



The rated current of this generated is 50 A, so 40 A is 80% of full load. If the full load armature reaction is 300 A·turns, and if the armature reaction is assumed to change linearly with armature current, then the armature reaction will be 240 A·turns. The demagnetizing effect of armature reaction is equivalent to a reduction in field current of

$$\Delta I_F = \frac{240 \text{ A•t}}{1000 \text{ t}} = 0.24 \text{ A}$$

The figure below shows that a triangle consisting of 7.2 V and (240 A·turns)/(1000 turns) = 0.24 A fits exactly between the E_A and V_T lines at a terminal voltage of 105 V.

Note that decreasing the field resistance of the shunt generator increases the terminal voltage.

8-27. A 120-V 50-A cumulatively compounded dc generator has the following characteristics:

$R_A + R_S = 0.21\ \Omega$ $\qquad$ $N_F = 1000$ turns

$R_F = 20\ \Omega$ $\qquad$ $N_{SE} = 25$ turns

$R_{adj} = 0$ to $30\ \Omega$, set to $10\ \Omega$ $\qquad$ $n_m = 1800$ r / min

The machine has the magnetization curve shown in Figure P8-7. Its equivalent circuit is shown in Figure P8-10. Answer the following questions about this machine, assuming no armature reaction.

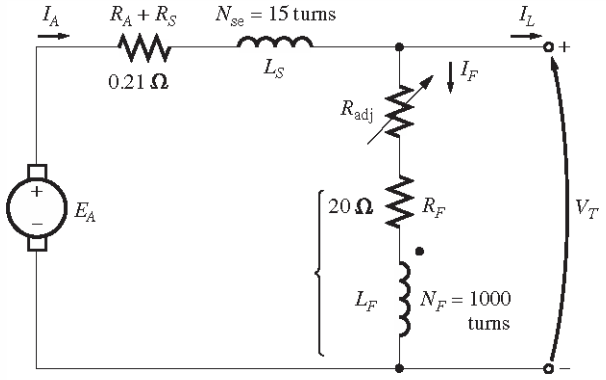


FIGURE P8–10
The compounded dc generator in Problems 8–27 and 8–28.

(a) If the generator is operating at no load, what is its terminal voltage?

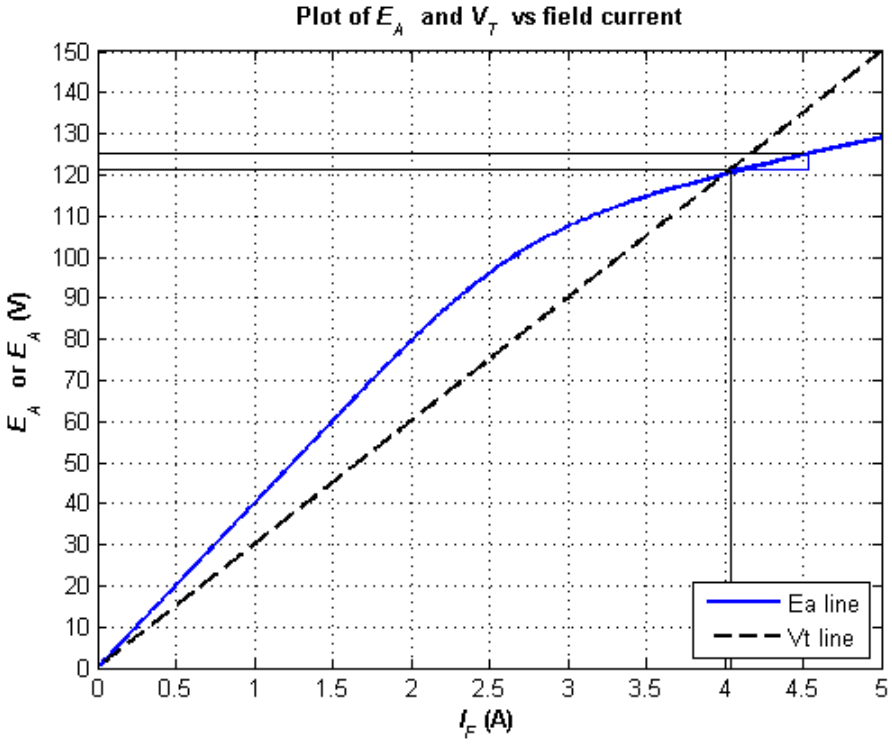
(b) If the generator has an armature current of 20 A, what is its terminal voltage?

(c) If the generator has an armature current of 40 A, what is its terminal voltage'?

(d) Calculate and plot the terminal characteristic of this machine.

SOLUTION

(a) The total field resistance of this generator is 30 Ω, and the no-load terminal voltage can be found from the intersection of the resistance line with the magnetization curve for this generator. The magnetization curve and the field resistance line are plotted below. As you can see, they intersect at a terminal voltage of 121 V.



(c) If the armature current is 40 A, then the effective field current contribution from the armature current

$$\frac{N_{\text{SE}}}{N_F} I_A = \frac{25}{1000}(40\text{ A}) = 1.0\text{ A}$$

and the $I_A(R_A + R_S)$ voltage drop is $I_A(R_A + R_S) = (40\text{ A})\ (0.21\,\Omega) = 8.4\text{ V}$. The location where the triangle formed by $\frac{N_{\text{SE}}}{N_F} I_A$ and $I_A R_A$ exactly fits between the E_A and V_T lines corresponds to a terminal voltage of 121 V, as shown below.